Balancing weights against maximum likelihood propensity scores: a simulation study of `balnet` across sample size, overlap, dimension and penalty choice

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Abstract

Maximum likelihood fits propensity scores to predict treatment, balancing weights to balance covariates directly. `balnet` fits balancing weights by a lasso-penalised balancing loss whose path runs to exact balance or stops short where infeasible. We compare `balnet` with cross-validated lasso MLE weights, alone and augmented, by simulation across sample size, overlap, dimension past $p = n$, confounder density and outcome signal, with and without misspecification. Balancing weights have lower RMSE throughout, by a third to two thirds on average, except noisy outcomes and the poorest overlap. Whether the path reaches exact balance decides what else helps. Augmentation adds nothing there and restores the lead where it stops short. A second study finds stopping inside the path lowers RMSE for noisy outcomes or, once augmented, limited overlap. Cross-validating the balancing loss captures this most often, though none is best in more than three cells in ten. We recommend augmented balancing weights tuned this way.

Keywords: balancing weights; propensity score; inverse probability weighting; lasso; Monte Carlo simulation.

1 Introduction

Much of the evidence on programmes, treatments and interventions is observational. Without randomisation, the groups compared differ in their pre-treatment characteristics, and those differences, not the treatment, may explain any gap in outcomes.

The standard approach is to assume conditional independence, meaning that once the relevant pre-treatment characteristics are controlled for, treatment allocation is as good as random. Key to this is the propensity score, the probability of receiving treatment given the observed covariates (Rosenbaum and Rubin 1983). One use of the score is inverse-propensity weighting (IPW), which weights each unit by the inverse of its estimated probability of the treatment it received. In medical research its use has increased rapidly, for example in comparing treatments from hospital records where patients were never randomised (Austin and Stuart 2015; Ben-Michael and Keele 2023). In economics, IPW is used to evaluate programmes such as job training, where the effect on participants' earnings must be recovered from a non-experimental comparison group (Busso et al. 2014).

In practice the score is unknown and must be estimated. The usual estimator is logistic regression fit by maximum likelihood (MLE). It chooses the parameters of the propensity model so that the

fitted score predicts the observed treatment assignment as well as possible (Wager 2024). The goal of weighting here however, is not to predict treatment assignment accurately but to balance covariates for comparable treatment groups (Zhao 2019).

While IPW with the oracle propensity score balances covariate means exactly, MLE-estimated propensity scores do not generally achieve this, even without model misspecification (Wager 2024; Sverdrup and Hastie 2026). Under misspecification or in small samples, small estimation errors are amplified by inversion of the propensity score and can produce severe imbalance and biased estimates (Ben-Michael et al. 2021). This problem is compounded by weight instability, as observations with small propensity scores receive disproportionately large weights, inflating variance and leading researchers to ad hoc solutions (Chattopadhyay et al. 2020). Furthermore, these fixes presume a balance check that in a review of applied medical studies using IPW, most did not make (Austin and Stuart 2015).

Instead of modelling the score, covariate balancing methods choose the weights to target directly the balance that IPW requires. The covariate balancing propensity score (CBPS) applies this within a propensity model, selecting its parameters to satisfy the sample balance conditions (Imai and Ratkovic 2014). As a result, their estimates tend to be more robust to model misspecification and more efficient than those from MLE weights (Chattopadhyay et al. 2020). Under a correctly specified linear outcome and logistic propensity model, this efficiency is exact and IPW with balancing weights is $\sqrt{n}$-consistent with lower variance than even the oracle IPW (Wager 2024).

The theoretical case for covariate balancing is well established, but most implementations struggle with large or high-dimensional data. CBPS and entropy balancing approaches both enforce exact balance conditions, which become unworkable in large-scale data settings (Fong et al. 2022; Sverdrup and Hastie 2026). Stable balancing weights from Zubizarreta (2015) require a maximum imbalance tolerance to be set before estimation and must be re-solved if this tolerance is unattainable or the bias-variance tradeoff is unattractive (Sverdrup and Hastie 2026). Zhao (2019) shows balancing estimators can generally be expressed as minimising a covariate balancing loss function, which opens them to lasso and other penalties. RCAL from Tan (2020b) implements lasso regularisation for a covariate balancing loss function, but as the package is written in R it struggles computationally with large data.

Like balancing approaches, doubly robust estimators offer protection against misspecification. These estimators, referred to as augmented estimators, combine the weights with an outcome model and remain consistent if either the propensity model or the outcome model is correctly specified (Ding 2024). The standard version is the augmented IPW (AIPW) estimator. It corrects the regression estimate by applying the IPW weights to the outcome model's residuals and adding their average (Robins et al. 1994). Where exact balance is out of reach, AIPW has outperformed unaugmented balancing weights (Athey et al. 2018).

Much recent work builds doubly robust estimators for the balancing approach, many obtaining the weights by optimising for balance directly rather than by first estimating a propensity score (Wang and Zubizarreta 2020; Hirshberg and Wager 2021; Bruns-Smith et al. 2026). These augmented estimators therefore sit outside the class of covariate balancing propensity scores that Zhao (2019) unifies. Within that class, Tan's augmented estimator fits the outcome model with weights from

the same propensity score, so changing the weights changes the outcome model too (Tan 2020a).[1] Bruns-Smith et al. (2026) discuss a general augmented balancing weight (ABW) estimator, where the outcome model is held fixed and the IPW weights are replaced by balancing weights. Here, when both models are linear in the same covariates, ABW is a single regression combining the outcome model and OLS, which collapses to OLS at exact balance (Wager 2024). For weights from the covariate balancing loss above, which come through a logistic link, this collapse holds only up to an approximation error.

Of the balancing implementations above, `balnet` from Sverdrup and Hastie (2026) applies modern optimisation techniques to the regularised covariate balancing loss. Like RCAL it uses lasso regularisation, but through a core solver written in C++ that scales to large and high-dimensional data. The solver also supports ridge and elastic net penalties, which Study 2 uses. Its path for $n = 100{,}000$ and $p = 500$ computes in under a minute, while alternative balancing packages failed to produce results within an hour. Under lasso regularisation the penalty parameter λ is equivalent to the maximum allowable covariate imbalance, so its regularisation path provides a practical diagnostic for overlap and lets the analyst set the imbalance to tolerate (Sverdrup and Hastie 2026). The path ends at the target imbalance rather than at exact balance, and truncates earlier where that target is infeasible. The imbalance left over can then be handed to an outcome model (Wager 2024).

For modelling approaches it is standard practice to choose λ by cross-validation (CV) of the prediction error, a choice one would reasonably expect to extend to balancing loss approaches.[2] However, for balancing weights recent work cautions against applying cross-validation naively, finding that no tuning rule is uniformly best (Bruns-Smith et al. 2026).

For balancing methods the held-out loss becomes the balancing loss or the remaining covariate imbalance. CV then assumes the held-out loss is an unbiased estimate of the error in the estimated effect, but the penalty trades observable variance against unobservable bias, a trade CV cannot reliably make (Tan 2020b; Bruns-Smith et al. 2026; Athey et al. 2018). Following the reasoning behind balancing weights, Wang and Zubizarreta (2020) score balance on bootstrap samples over a grid of approximate balance tolerances and find that under poor overlap exact balance is not always optimal. On the `balnet` path this becomes a question of where to set λ, or whether to set it at all, as by construction its truncated path already produces approximate balancing weights when exact balance is not feasible.

Study 1 asks to what degree and when balancing weights improve on MLE weights, whether used alone or in an augmented estimator. Study 2 then asks where `balnet`'s path endpoint does not minimise root mean squared error (RMSE), and whether at those points any selector beats it. Across both studies the dividing line is whether the `balnet` path reaches exact balance, which with the outcome signal tells the analyst when balancing weights, augmentation and a selector each pay.

[1] Tan's estimator refits the outcome model whenever the weights change, so a comparison between weights would also be a comparison between outcome models. Applying it here would confound Study 1, which compares across weighting estimators.

[2] Both RCAL and `balnet` provide a CV setting with tunable folds in their base package.

2 Aims

2.1 Study 1

Our first study aims to guide the choice between balancing weights and MLE weights, whether used alone or in an augmented estimator. The settings that follow vary the sample size, overlap, dimension, the number and correlation of confounders, and the strength of the outcome signal. Each is run under both a correctly specified and a misspecified propensity score. Following Tan (2020b), we focus on how the two sets of weights perform relative to each other across these settings and over three outcome functions.

Both sets of weights are also augmented with the same lasso outcome model, so ABW and AIPW differ only in their weights. Comparing the two shows whether differences between balancing and MLE weights hold once an outcome model is added. At exact balance, $\texttt{balnet}_{10^{-4}}$ differs from OLS by the approximation error of its logistic weights (Bruns-Smith et al. 2026). We track this error where OLS can be fitted and the path reaches its target.

Sample size: Under a correctly specified propensity score both approaches are consistent (Wager 2024). For a linear outcome, IPW with balancing weights is expected to have lower variance than even IPW with the true propensity score. Here we ask how the sample size affects the relative performance of $\texttt{balnet}_{10^{-4}}$ and `cv.glmnet`, a baseline for the later comparisons.

Under a misspecified propensity score, IPW estimators can converge away from the true estimand, and with MLE weights their bias and RMSE have been found to worsen as n grows (Waernbaum and Pazzagli 2023; Kang and Schafer 2007). We therefore ask whether the bias of `cv.glmnet` settles as n rises to 10^6, and how $\texttt{balnet}_{10^{-4}}$ compares. We also track AIPW, which has been found to compound errors when both models are wrong (Kang and Schafer 2007). Where a bias settles while the sampling error shrinks, coverage must collapse with n, which we report at the empirical SE (Morris et al. 2019).

Overlap: We then compare the two sets of weights as overlap degrades, meaning that treated and control units increasingly lack counterparts with similar propensity scores. We also include levels at which overlap is practically violated, as defined in the data-generating mechanism.

With a correctly specified outcome model, AIPW has outperformed unaugmented approximate balancing weights at both overlap levels studied (Athey et al. 2018). Under poor overlap and a misspecified outcome model, however, unaugmented approximate balancing weights have produced lower RMSE than AIPW (Chattopadhyay et al. 2020). Besides these swapped signs, we ask how $\texttt{balnet}_{10^{-4}}$ performs once exact balance is out of reach, as exact balancing weights can become infeasible under poor overlap (Hainmueller 2012; Sverdrup and Hastie 2026).

High-dimensional settings: In healthcare especially, propensity scores are fitted to a large number of covariates, as in database studies (Wyss et al. 2026). By raising the number of covariates to $p/n = 4$ we cover these settings, which force the MLE weights to select four confounders within thousands of covariates and the balancing weights to balance thousands. Here we also ask how the balancing loss behaves as p grows.

Firstly, exact balance is expected to fail for $\texttt{balnet}_{10^{-4}}$ around $p/n = 0.5$, so that its path stops short of its target. Here, we look for the error of `cv.glmnet` to grow gradually with p (Tan 2020a),

and for ABW to separate from $\texttt{balnet}_{10^{-4}}$ (Wager 2024). Against AIPW, augmented balancing weights have kept a lead in past work, though not one that consistently widened with p (Athey et al. 2018; Ning et al. 2020), so we also examine this lead.

Secondly, we ask whether misspecification changes how the modelling estimators respond to added covariates. Under misspecification, the RMSE of IPW with MLE weights fell as p grew in Tan (2017) but rose in Athey et al. (2018). Running both specifications on the same draws lets us consider the effects of p and misspecification separately.

With a denser covariate profile: The p/n grid is limited to four confounders, as Kang–Schafer misspecification requires (Kang and Schafer 2007). This is restrictive, as conditional independence typically needs many covariates to be plausible. Here, we increase the number of confounders s to 16 and 64, allow their signal to even out as the decay rate ν falls from 1 to 0.5, and consider independent, AR(1)-correlated and equicorrelated covariance structures. Each design also has a misspecified propensity score, built from the transformation of Tan (2020a). There, balancing weights from the same loss as `balnet` have substantially outperformed MLE weights (Tan 2020b), which we test at every s.

At $p/n = 0.1$, the error from modelling weights is bounded at a rate that grows with the number of active covariates (Tan 2020a). In contrast, exact balancing weights give an unbiased estimate for a linear outcome at any s (Ben-Michael et al. 2021). The advantage of balancing over modelling weights should then widen as s rises. From $p/n = 0.5$, however, exact balance should fail, and the bias of balancing weights for a linear outcome is then at most the largest remaining imbalance scaled by the ℓ_1 norm of the outcome coefficients (Zhao 2019). As the outcome signal has fixed variance, spreading it over more confounders raises this norm, so the bias should rise with s and as ν falls, except under equicorrelation. Correlation may pull the other way. When the outcome variance is held fixed, correlated confounders absorb part of each other's signal and their coefficients shrink, lowering the ℓ_1 norm that bounds the bias, most of all under equicorrelation.

Raising s also makes the propensity model denser, which matters for the augmented estimators. AIPW relies on a sufficiently sparse propensity model, whereas augmented balancing weights assume nothing about treatment assignment beyond overlap (Athey et al. 2018). AIPW should then lose ground to ABW as that model grows denser.

Outcome signal: Lastly, we ask how the strength of the outcome signal changes the comparison between balancing and modelling weights. Here, the outcome signal strength κ rises from 0.02 to 1 against a fixed noise level, under both correctly specified and misspecified propensity scores.

As κ falls, the bias that exact balance removes shrinks, but its cost in variance does not. At low κ, $\texttt{balnet}_{10^{-4}}$ may then lose ground to modelling weights. At larger λ its path gives approximate balance, which lets us choose how much bias to trade for variance, though the best trade depends on the signal-to-noise ratio (Wang and Zubizarreta 2020; Zhao 2019).

2.2 Study 2

In preliminary runs a cross-validated λ beat the path endpoint for IPW only when the outcome signal was weak. Our second study therefore asks whether tuning or bootstrapping approaches can offer a better stopping point than the endpoint. First, when does the RMSE-minimising λ come

before the path endpoint, an interior optimum. Second, where and how often does one of our five selectors improve on the endpoint, a selector win.

By construction no selector sees the outcome, so each picks the λ on the path that minimises one of three criteria. The first, `bloss`, is the balancing loss that `balnet` minimises along its path. The second, `smd`, is the mean absolute standardised mean difference across covariates after weighting, the bootstrap balance measure of Wang and Zubizarreta (2020) averaged rather than summed. The third, `inf`, is the largest absolute standardised mean difference after weighting, the quantity our λ bounds under lasso.

Each criterion is scored by cross-validation or by a half-sample bootstrap motivated by Wang and Zubizarreta (2020). The balancing loss is scored by cross-validation only, giving five selectors, as its fit minimises its own target and scoring it on sub-samples would favour the endpoint deterministically, analogous to scoring a likelihood at its own maximiser.

We are especially interested in overlap and outcome signal. Approximate balance has been found to lower RMSE relative to exact balance when overlap is limited (Wang and Zubizarreta 2020). For augmented balancing weights with ridge penalties, the RMSE-minimising penalty has been found to grow with outcome noise, which selectors using only the covariates and treatment cannot see (Bruns-Smith et al. 2026). If this holds for `balnet`, interior optima and selector wins should both become more likely as κ falls. We also ask whether moving the penalty form α from lasso towards ridge lowers RMSE, at the endpoint or at the interior optimum.

3 Methods

3.1 Estimand

Our estimand is the average treatment effect, $\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$. We set $\tau = 0$ in every cell so $\hat{\tau}$ is its own error by construction.

3.2 Data-generating mechanism

Simulation design: We combine data-generating process (DGP) designs from the IPW literature into one model that reproduces the stress conditions common in observational data. The design of Kang and Schafer (2007), extended to high-dimensional settings by Tan (2020b), provides our baseline for testing misspecification. As overlap is a key assumption of IPW, an overlap scaler on the propensity score gives a range of overlap conditions (Busso et al. 2014; Keele et al. 2026). While the design of Tan (2020b) is high-dimensional, only four of its covariates are active. A sparsity parameter, as in Zhao (2019), allows the more realistic case where many covariates are active. Autoregressive covariance structures, as in Tan (2020a) and Zhao (2019), allow simulation under moderate correlation, and an equicorrelated structure extends that correlation to every covariate. Finally, since a comparison of RMSE depends on the sample size, we vary it (Morris et al. 2019).

Within this design, each cell draws n observations of p multivariate normal covariates $X_i = (X_{i1}, \dots, X_{ip})$ with mean zero and unit variance. Their covariance takes three levels. Setting $\text{cov}(X_{ij}, X_{ik}) = \rho^{|j-k|}$ gives iid covariates at $\rho = 0$ and an AR(1) structure at $\rho = 0.5$, and a third, equicorrelated level sets $\text{cov}(X_{ij}, X_{ik}) = 0.5$ for all $j \neq k$ through one common factor, $X_{ij} = \sqrt{0.5}\, Z_{i0} + \sqrt{0.5}\, Z_{ij}$ with $Z_{i0}, Z_{ij} \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$ (Friedman et al. 2010). Under AR(1)

the active covariates are adjacent, which may limit the effect of correlation (Athey et al. 2018), whereas under equicorrelation every inactive covariate shares the common factor with the active ones. The treated proportion is 0.5, as in Kang and Schafer (2007).

The true propensity score is

$$e^*(X_i) = \left\{1 + \exp\left(\frac{g_i}{c\,\sigma_g} - \alpha\right)\right\}^{-1}, \qquad g_i = \sum_{j=1}^{s} j^{-\nu} X_{ij}, \tag{1}$$

where s is the number of active covariates, ν is their signal decay rate along j and α sets the treated proportion. Overlap holds for any overlap scaler $c > 0$, with $c = 1$ as its base to roughly correspond with Tan (2020b).[3] Lowering c weakens overlap by pushing $e^*(X_i)$ towards 0 and 1.

Appendix Table 6 shows the proportion of units whose true propensity score falls outside [0.01, 0.99] and [0.05, 0.95] at each level of c, and Appendix Figure 1 displays the density of $e^*(X_i)$ for each arm. In the terms of Wager (2024), our design satisfies weak but not strong overlap, as $e^*(X_i)$ stays strictly between 0 and 1 at any $c > 0$ but is never bounded away from them. Under sufficiently degraded overlap, samples contain covariate regions with only treated or only control units, which we consider practical violations of common support following Chattopadhyay et al. (2020). Such violations are possible at $c = 0.5$, likely at 0.25 and almost certain by $c \leq 0.05$.

To prevent changes in s, ν and the covariance structure shifting overlap, we standardise g_i using its population variance $\sigma_g^2 = \sum_{j,k \leq s} (jk)^{-\nu} \operatorname{cov}(X_{ij}, X_{ik})$. Dividing by σ_g makes g_i/σ_g standard normal, leaving only c and α to shift $e^*(X_i)$. To prevent c shifting the treated proportion, we select α numerically so that the expected value of $e^*(X_i)$ equals the target, which gives $\alpha = 0$ at 0.5[4]. Treatment is assigned as $W_i \sim \text{Bernoulli}\{e^*(X_i)\}$. Potential outcomes are generated as $Y_i(0) = Y_i(1) = \kappa\, h(X_i) + \varepsilon_i$, where $\varepsilon_i \sim \mathcal{N}(0, 1)$ and κ sets the outcome signal strength, so at $\kappa = 0$ the outcome is noise alone. Outcome specifications follow Tan (2020b) and use the first four covariates, the active set when $s = 4$,

$$\begin{aligned} h^{\text{lin}}(X_i) &= X_{i1} + 0.5\,X_{i2} + 0.5\,X_{i3} + 0.5\,X_{i4} \\ h^{\text{quad}}(X_i) &= \sum_{j=1}^{4} \{\max(X_{ij}, 0)\}^2 \\ h^{\text{exp}}(X_i) &= \sum_{j=1}^{4} \exp(X_{ij}/2) \end{aligned} \tag{2}$$

In the sparsity grid, to keep every active covariate a confounder as s varies, we instead let h load on all s active covariates, weighted by $j^{-\nu}$.

Misspecification structure: We force misspecification with the covariate transformations of Kang and Schafer (2007), in which the estimators observe transformed covariates $\tilde{X}$ rather than the true

[3] The design in Tan (2020b) corresponds to $c = 1/1.15 \approx 0.87$ so our base case has slightly stronger overlap levels, due to our standardisation.

[4] Since g_i/σ_g is standard normal, this mean depends only on α and c and rises strictly in α, so the solution is unique and found by `uniroot`.

X,

$$\begin{aligned}
\tilde{X}_{i1} &= \exp(X_{i1}/2) \\
\tilde{X}_{i2} &= \frac{X_{i2}}{1+\exp(X_{i1})} + 10 \\
\tilde{X}_{i3} &= (0.04\, X_{i1}\, X_{i3} + 0.6)^3 \\
\tilde{X}_{i4} &= (X_{i2} + X_{i4} + 20)^2.
\end{aligned} \tag{3}$$

As in Tan (2020b), $\tilde{X}_{i1}, \ldots, \tilde{X}_{i4}$ are standardised within each sample to mean zero and unit variance, and the remaining covariates are observed as they are. The transformations are written for four covariates, so we apply them at $s = 4$, where the transformed covariates are exactly the confounders shared by $e^*(X_i)$ and Y_i.

These transformations misspecify both models our augmented estimators fit, a logistic score and a linear outcome model. Here, the data-generating score and outcome do not change. They are still a logit and a linear regression in X, and written in the observed $\tilde{X}$ they remain linear in the parameters with nonlinear terms in $\tilde{X}$. The estimators are misspecified only in that they enter the transformed covariates themselves as the linear terms. This gives a realistic misspecification structure, in which a slight error in a researcher's functional form is enough to misspecify a model (Kang and Schafer 2007).

Table 1: Specification of the propensity score and the outcome model in each arm of the design, by outcome.

Arm	$\hat{e}$ on	Outcome on	$\hat{e}$	Outcome linear	Outcome nonlinear
Correct specification	X	X	correct	correct	misspecified
Misspecification	$\tilde{X}$	$\tilde{X}$	misspecified	misspecified	misspecified
Refit on X (`_ortrue`)	$\tilde{X}$	X	misspecified	correct	misspecified

Only the augmented estimators have an outcome model, so only they reach the third row of Table 1. There the outcome model is refitted on X and is correct, so its residuals have mean zero and weighting them with a misspecified score adds no bias. The second row is the case Kang and Schafer (2007) describe as the practical motivation behind doubly robust estimators, where a practitioner naturally assumes a linear outcome model and a logistic score are adequate because both look trustworthy. There, naive application of doubly robust estimators can give errors that grow with sample size, which we call double fragility, following Testa et al. (2026).

Beyond $s = 4$ the Kang–Schafer transformations no longer reach every confounder, so our denser grids use the transformation of Tan (2020a) instead. Here, $X_{ij}^{\dagger} = X_{ij} + \{\max(X_{ij} + 1, 0)\}^2$ is applied to every active covariate and standardised within each sample, while W_i and Y_i are still generated from X. To keep the same practical interpretation of misspecification, we deviate from Tan (2020a) by placing $X^{\dagger}$ in the observed covariates rather than the true models. Both models fitted on $X^{\dagger}$ are then misspecified, as before.

Simulation settings: Each grid varies one or more factors against a base case of $n = 1000$, $p = 50$, $s = 4$, $\nu = 1$, $\rho = 0$, $c = 1$, $\kappa = 1$ and a treated proportion of 0.5, with all three outcomes computed on each draw. Most grids vary one factor (Table 2). Sample size is crossed with overlap, dimension, path endpoint and treated proportion, and the denser grids cross five factors, a non-factorial structure chosen for computational time (Morris et al. 2019). Within a grid every

estimator and both specifications run on the same draws, from L'Ecuyer–CMRG streams indexed by cell and replication, so only the observed covariates differ between the correct and misspecified arms. Cells use 10 000 replications, except 1000 on the $n \times c$ and denser grids, with Monte Carlo standard errors on every measure, and a replication in which an estimator fails is kept as missing and counted (Morris et al. 2019). All runs use R 4.5 or 4.6 (R Core Team 2025) with `balnet` 0.0.4 (Sverdrup and Hastie 2026) and `glmnet` 5.0 (Friedman et al. 2010). The code, run scripts and stored summaries are available from the author.

Table 2: The Study 1 grids.

Grid	Factor varied	Levels	Specification	Replications
Sample size	n	{500, 1k, 5k, 10k, 50k, 100k, 200k, 400k}	correct	10 000
Sample size	n	{500, 1k, 5k, 10k, 50k}; 250 to 200k (29) and 1M	misspecified	10 000
Overlap	c, at n = 1000 and 10 000	{0.05, 0.15, 0.25, 0.5, 0.75, 1, 2}	both	10 000
Sample size × overlap	n, c	100 to 5000 (12) × 0.01 to 1 (13)	both	1000
Dimension	p/n, at n = 500 and 1000	{0.05, 0.1, 0.5, 1, 2, 4}	both	10 000
Denser profile	n, p/n, s, ν, ρ	{500, 1000} × {0.1, 0.5, 1} × {4, 16, 64} × {1, 0.5} × {0, 0.5}, $s \leq p$	both	1000
Equicorrelated	as above at cov = 0.5	34 cells	both	1000
Outcome signal	κ	{0.02, 0.05, 0.1, 0.2, 0.35, 0.5, 0.7, 1}	both	10 000
Treated proportion	treated proportion, at n = 1000 and 10 000	{0.05, 0.1, 0.2, 0.5}	both	10 000
Path endpoint	n, maximum imbalance	{500, 1k, 5k, 10k, 50k} × 10^{-2} to 10^{-7} by half decades (11)	both	10 000

A range gives its number of levels in brackets. Factors not listed take their base-case values. The treated proportion and path endpoint grids were run but are not reported.

3.2.1 Study 2

Study 2 changes two things in the design of Study 1. Here, each replication carries seven outcomes on one draw, with κ halved from one to $1/32$ and then set to zero, so the selectors, folds and half-samples are shared along the noise axis. The penalty form α runs from ridge through the elastic net to lasso. Overlap takes six of the seven values of Study 1, without $c = 2$. Sample size, dimension and misspecification are each crossed with overlap, and n and p with α as well. A denser grid takes the confounder profile $s \in \{4, 16, 64\}$ under its three covariance structures, at $n = 1000$, $p = 100$ and $c = 0.25$ under both specifications. Every grid runs 2500 replications except the two largest samples at 1000 (Table 3).

3.3 Methodology

We consider the standard potential outcomes setting with a sample of n units, a treatment indicator $W_i \in \{0, 1\}$ and potential outcomes $Y_i(1)$ and $Y_i(0)$ under treatment and control. Identification of the ATE, $\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$, relies on conditional independence and overlap. Under these assumptions the propensity score $e(x) = P[W_i = 1 \mid X_i = x]$ allows unbiased estimation via IPW:

$$\tau = \mathbb{E}\left[\frac{W_i Y_i}{e(X_i)} - \frac{(1 - W_i)Y_i}{1 - e(X_i)}\right]. \tag{4}$$

Propensity score estimation: We compare two approaches under a logistic propensity score model,

Table 3: The Study 2 grids.

Grid	Factor varied	Levels	Specification	Replications
Overlap	c	{0.05, 0.15, 0.25, 0.5, 0.75, 1}	correct	2500
Outcome noise	κ	{1, 1/2, 1/4, 1/8, 1/16, 1/32, 0}	correct	2500
Noise × overlap	κ, c	{1, …, 1/32, 0} × {1, 0.5, 0.25, 0.15}	correct	2500
Penalty form × overlap	α, c	{0, 0.25, 0.5, 0.75, 1} × {1, 0.5, 0.25, 0.15}	correct	2500
Sample size × overlap × penalty	n, c, α	{250, 500, 1000, 2000} × {0.25, 0.15} × {0.5, 1}	correct	2500
Large n × overlap	n, c	{5000, 10 000} × {0.25, 0.15}	correct	1000
Dimension × overlap × penalty	p, c, α	{50, 250, 500, 1000} × {0.5, 0.25} × {0.5, 1}	correct	2500
Misspecification × overlap	c	{0.05, 0.15, 0.25, 0.5, 0.75, 1}	misspecified	2500
Denser profile	s, covariance	{4, 16, 64} × {iid, AR(1), equi}, at p = 100, ν = 0.5, c = 0.25	both	2500

The base case is lasso at n = 1000, p = 50, s = 4 and c = 1, with four confounders among iid covariates; factors not listed take these values. From the penalty-form grid onward every cell carries all seven κ levels on one draw.

$e(x) = 1/[1 + \exp\{-(\beta_0 + x^\top \beta)\}]$. Both use penalised logistic regression but differ in the loss function minimised.

Standard MLE approach: The MLE approach estimates β by minimising the negative Bernoulli log-likelihood,

$$\ell_{\text{ML}}(\beta) = -\frac{1}{n}\sum_{i=1}^{n}\left[W_i\eta_i - \log\{1+\exp(\eta_i)\}\right], \tag{5}$$

where $\eta_i = \beta_0 + X_i^\top \beta$. This targets accurate prediction of treatment assignment, with first-order conditions

$$\frac{1}{n}\sum_{i=1}^{n}\{W_i - \hat{e}(X_i)\}X_{ij} = 0, \quad j = 1, \dots, p, \tag{6}$$

that balance predicted and observed treatment whatever the specification. We use the penalised MLE of the `glmnet` package (Friedman et al. 2010), which computes a regularisation path in a penalty parameter λ that acts as a coefficient budget, with lasso penalties and λ chosen by five-fold cross-validation at its minimum-deviance value, the estimator we write `cv.glmnet`.

Covariate balancing approach: The covariate balancing approach starts from the finite-sample bias of IPW estimators, which is driven by covariate imbalance (Wager 2024, ch. 7). Achieving this balance in finite samples requires

$$\frac{1}{n}\sum_{i=1}^{n}\frac{W_i X_{ij}}{\hat{e}(X_i)} = \frac{1}{n}\sum_{i=1}^{n} X_{ij}, \quad j = 1, \dots, p. \tag{7}$$

For estimating treatment effects the covariate balancing approach uses two separate fits, targeting balance for the treated and control means. When estimating the treated mean $\mathbb{E}[Y_i(1)]$, this balance condition is enforced by a loss function with first-order conditions

$$\frac{1}{n}\sum_{i=1}^{n}\left(\frac{W_i}{\hat{e}(X_i)} - 1\right)X_{ij} = 0, \quad j = 1, \dots, p, \tag{8}$$

which integrates to the covariate balancing loss function

$$\ell_1(\eta) = \frac{1}{n}\sum_{i=1}^{n}\left[W_i \exp(-\eta_i) + (1 - W_i)\eta_i\right]. \tag{9}$$

The corresponding loss function $\ell_0(\eta)$ for the control mean $\mathbb{E}[Y_i(0)]$ follows by symmetry. For exact balancing, the condition for a solution to exist is also stricter under this approach than with MLE, requiring that no linear combination of the covariates is non-negative on every treated unit while averaging at or below zero over the controls (Tan 2020b). To address this, we regularise the covariate balancing loss function using the penalised approach of Sverdrup and Hastie (2026), where the penalised objective under lasso regularisation is

$$\ell_1(\eta) + \lambda \sum_{j=1}^{p} |\beta_j|. \tag{10}$$

Under lasso penalisation, the exact balance requirements from Equation 7 are relaxed by the KKT conditions giving

$$\left| \frac{1}{n} \sum_{i=1}^{n} \frac{W_i X_{ij}}{\hat{e}(X_i)} - \frac{1}{n} \sum_{i=1}^{n} X_{ij} \right| \leq \lambda, \quad j = 1, \ldots, p. \tag{11}$$

Here λ is the maximum allowable imbalance across covariates, in standardised mean differences since `balnet` standardises the covariates by default. The regularisation path then gives a sequence of solutions at decreasing imbalance tolerances, which also serves as a diagnostic for overlap (Sverdrup and Hastie 2026). When further reductions in imbalance are infeasible, `balnet` truncates its path before the path floor $\lambda_{min} = 10^{-4}$, and we count a replication as truncated when either arm's path ends above it. The estimator at the path end is written $\texttt{balnet}_{10^{-4}}$. In the terms of Zhao (2019), `balnet` is then an approximate balancing estimator, defined by the imbalance left in each arm, bounded by λ.[5]

Modelling IPW estimators: Modelling aims for covariate balance by predicting the propensity score, on the intuition that a correctly estimated propensity score will balance confounders through IPW estimation.

1. The Horvitz–Thompson (HT) IPW estimator for the ATE,

$$\hat{\tau}_{\text{HT}} = \frac{1}{n} \sum_{i=1}^{n} \left[\frac{W_i Y_i}{\hat{e}(X_i)} - \frac{(1 - W_i) Y_i}{1 - \hat{e}(X_i)} \right], \tag{12}$$

   weights each unit by the inverse of its estimated propensity score $\hat{e}(X_i)$, following the survey sampling estimator from Horvitz and Thompson (1952). Under the oracle propensity score it is unbiased for τ (Wager 2024, ch. 2).

2. The Hájek (H) estimator,

$$\hat{\tau}_{\text{H}} = \frac{\sum_{i=1}^{n} W_i Y_i / \hat{e}(X_i)}{\sum_{i=1}^{n} W_i / \hat{e}(X_i)} - \frac{\sum_{i=1}^{n} (1 - W_i) Y_i / \{1 - \hat{e}(X_i)\}}{\sum_{i=1}^{n} (1 - W_i) / \{1 - \hat{e}(X_i)\}}, \tag{13}$$

   normalises the HT estimator by dividing both groups by the sum of their weights, bounding the estimate by the observed outcomes to avoid minor errors in $\hat{e}(X_i)$ causing large errors in $\hat{\tau}$ through inversion (Hájek 1971; Ding 2024, ch. 11).

[5] For $\alpha < 1$, used in Study 2, each covariate's imbalance is no longer capped and grows with its coefficient, moving `balnet` from the lasso's control of the worst covariate toward ℓ_2 ridge balancing (Bruns-Smith et al. 2026).

For the balancing loss approach there is no distinction between HT and H, as the weights in each arm sum to n by construction when the balancing loss is minimised (Tan 2020b; Sverdrup and Hastie 2026). Three references complete the set. The oracle is each estimator at the true propensity score, OLS is fitted within each arm on the observed covariates and predicted for all units (Kang and Schafer 2007), and the unadjusted estimator is the difference in means.

Doubly robust IPW estimators: To mitigate the effects of model misspecification, both doubly robust estimators estimate an outcome model alongside the propensity score, as only one need be correctly specified.

1. The augmented IPW (AIPW) estimator (Robins et al. 1994).

$$\begin{aligned}\hat{\tau}_{\text{AIPW}} = \frac{1}{n}\sum_{i=1}^{n}[\hat{m}_1(X_i) - \hat{m}_0(X_i)] \\ + \frac{\sum_i \hat{w}_i^{(1)}\{Y_i - \hat{m}_1(X_i)\}}{\sum_i \hat{w}_i^{(1)}} - \frac{\sum_i \hat{w}_i^{(0)}\{Y_i - \hat{m}_0(X_i)\}}{\sum_i \hat{w}_i^{(0)}},\end{aligned} \tag{14}$$

   adds a penalised outcome model to the MLE weights. The outcome model $\hat{m}_w(x) = \hat{\beta}_{w,0} + x^\top \hat{\beta}_w$ estimates $m_w(x) = \mathbb{E}[Y_i(w) \mid X_i = x]$, and is fit within arm w and predicted for all n units. Residual terms are normalised as in the Hájek estimator (Ding 2024). The population bias for $\mu_1 = \mathbb{E}[Y_i(1)]$ is the product of the two models' errors,

$$\text{Bias}(\hat{\mu}_1^{\text{AIPW}}) = \mathbb{E}\left[\frac{e(X) - \hat{e}(X)}{\hat{e}(X)}\{m_1(X) - \hat{m}_1(X)\}\right], \tag{15}$$

   so one correct model is enough. In finite samples the error is bounded by the product of the two fits' errors, so with both models misspecified neither factor is small and the errors can compound (Ding 2024).

2. The augmented balancing weights (ABW) estimator, in the general form of Bruns-Smith et al. (2026), is for the ATE

$$\begin{aligned}\hat{\tau}_{\text{ABW}} = \frac{1}{n}\sum_{i=1}^{n}[\hat{m}_1(X_i) - \hat{m}_0(X_i)] \\ + \frac{\sum_i \hat{\gamma}_i^{(1)}\{Y_i - \hat{m}_1(X_i)\}}{\sum_i \hat{\gamma}_i^{(1)}} - \frac{\sum_i \hat{\gamma}_i^{(0)}\{Y_i - \hat{m}_0(X_i)\}}{\sum_i \hat{\gamma}_i^{(0)}},\end{aligned} \tag{16}$$

   where $\hat{\gamma}_i^{(1)}, \hat{\gamma}_i^{(0)}$ are the `balnet` weights at the path endpoint and $\hat{m}_w(x) = \hat{\beta}_{w,0} + x^\top \hat{\beta}_w$ is a lasso outcome model fit within arm w through the cross-validated penalty parameter δ_w, predicted for all n units.

   Bruns-Smith et al. (2026) show that with weights and outcome model linear in the same covariates, the ABW estimator collapses to OLS at exact balance,

$$\hat{\tau}_{\text{ABW}} = \frac{1}{n}\sum_{i=1}^{n}[\hat{m}_1(X_i) - \hat{m}_0(X_i)], \qquad \lambda_1 = \lambda_0 = 0. \tag{17}$$

   Under lasso (ℓ_1, $\alpha = 1$) the collapse at exact balance is the same. Away from it, the weights

correct balance only for covariates whose raw imbalance exceeds λ, with $\hat{m}_w$ handling the rest, a closed form that holds only for uncorrelated covariates (Bruns-Smith et al. 2026, Prop. 5.1–5.2 and supp. App. E.3).

The balancing weights of `balnet` are built on a logistic propensity score, so at exact balance the logistic link leaves $\hat{\tau}_{\text{ABW}}$ an approximation error relative to OLS (Bruns-Smith et al. 2026, Remark 5 and supp. Prop. D.1).

3.4 Regularisation

We fit both approaches under the elastic net framing of Friedman et al. (2010). Here α sets the form of the penalty, ridge at $\alpha = 0$, lasso at $\alpha = 1$ and the elastic net anywhere between. The penalty shrinks coefficients, scaled by λ, as each fit minimises

$$\ell_1(\eta) + \lambda\Big[(1-\alpha)\tfrac{1}{2}\sum_{j=1}^{p}\beta_j^2 + \alpha\sum_{j=1}^{p}|\beta_j|\Big]. \tag{18}$$

Ridge shrinks every coefficient towards zero leaving all nonzero, the lasso sets some at exactly zero, and the elastic net does both in a mix set by α. Study 1 uses only the lasso penalty (ℓ_1), while Study 2 covers all three.

3.5 Performance measures

Bias: the average error of the estimates, Bias $= \bar{\tau} - \tau$, where $\bar{\tau} = \frac{1}{m}\sum_{r=1}^{m}\hat{\tau}_r$ is the mean of the estimates $\hat{\tau}_r$ over m replications. As $\tau = 0$, bias is $\bar{\tau}$.

EmpSE: the standard deviation of the estimates across replications, EmpSE $= \sqrt{\frac{1}{m-1}\sum_{r=1}^{m}(\hat{\tau}_r - \bar{\tau})^2}$, measuring precision (Morris et al. 2019).

RMSE: the root mean squared error, RMSE $= \sqrt{\frac{1}{m}\sum_{r=1}^{m}(\hat{\tau}_r - \tau)^2}$, combining bias and precision, as $\text{RMSE}^2 \approx \text{Bias}^2 + \text{EmpSE}^2$.

MCSE: the Monte Carlo standard error, the uncertainty in each measure from using a finite m, derived as in Morris et al. (2019). Bands and bars in the figures are 95% Monte Carlo intervals. For bias, it is $\text{EmpSE}/\sqrt{m}$, for RMSE it is $\text{SD}\big[(\hat{\tau}_r - \tau)^2\big]/(2\,\text{RMSE}\sqrt{m})$ by the delta method, and for EmpSE it is $\text{EmpSE}/\sqrt{2(m-1)}$.

Coverage at the empirical SE: the share C of replications in which $\hat{\tau}_r \pm 1.96\,\text{EmpSE}$ covers τ, with Monte Carlo standard error $\sqrt{C(1-C)/m}$. No variance estimator is implemented for the balancing weights, so EmpSE stands in for the estimated width. With the width exact and the estimates near normal, bias is the only source of under-coverage, so this is the coverage a correctly sized interval would reach. Coverage at the nominal 0.95, the level the interval claims, means the bias is small against the spread, and coverage falls with n wherever the bias shrinks more slowly than $\sqrt{1/n}$ (Morris et al. 2019).

Relative RMSE performance $\bar{R}(\Theta)$: To guide the choice between estimators, we introduce an average relative performance indicator $\bar{R}(\Theta)$. Here, estimator A is compared with estimator B over Θ, a set of cells c, where each c is a simulation setting under our DGP. So that cells with large RMSE do not dominate, the ratio R_c is averaged on the log scale, with its exponent taken for

$\bar{R}(\Theta)$,

$$R_c = \frac{\text{RMSE}_{A,c}}{\text{RMSE}_{B,c}}, \qquad \bar{R}(\Theta) = 100\Big\{1 - \exp\Big(\frac{1}{|\Theta|}\sum_{c\in\Theta}\log R_c\Big)\Big\}, \tag{19}$$

so that $\bar{R}(\Theta) > 0$ indicates that, over Θ, estimator A has on average $\bar{R}(\Theta)$ per cent lower RMSE than B.

Share of bias $\bar{s}_m(\Theta)$: To show whether differences in $\bar{R}(\Theta)$ are driven by bias or variance, for each estimator m we report its share of bias using squared RMSE (MSE),

$$s_{m,c} = \frac{\text{bias}^2_{m,c}}{\text{RMSE}^2_{m,c}}, \qquad \bar{s}_m(\Theta) = \frac{1}{|\Theta|}\sum_{c\in\Theta} s_{m,c}. \tag{20}$$

A cell counts as a win for A when $\log R_c$ lies more than three Monte Carlo standard errors below zero, and as a loss when it lies more than three above, so cells within three MCSE of zero are ties. By holding $\tau = 0$ fixed, comparing ratios, jointly considering $\bar{R}(\Theta)$ and $\bar{s}_m(\Theta)$, and using RMSE, which is on the same scale as EmpSE, we address several precautions raised by Morris et al. (2019).

3.5.1 Study 2

Selectors: Each selector picks a λ from one full `balnet` fit per replication, which also supplies the path endpoint. Cross-validation refits the path on ten folds, scores every λ on the held-out fold and takes the λ with the lowest mean score. The three cross-validated criteria are scored on the same folds, so they differ only in the criterion. The bootstrap does not refit. It scores the weights of the full fit on 500 random half-samples of the data and takes the λ with the lowest mean score, following Wang and Zubizarreta (2020). Selection is made separately in the treated and control fits, so a selector returns one λ per arm.

Evaluation: Each selected λ gives both balancing estimators from Study 1. Under lasso the path ends at a maximum imbalance of 10^{-4}, as in Study 1. Under ridge and the elastic net λ no longer bounds the standardised mean differences, so the path ends at a fixed fraction of λ_{max} instead. To locate the RMSE-minimising λ, both estimators are also evaluated at every point of a fixed λ grid per cell, set from a pilot draw and spaced evenly on the log scale from λ_{max} to the path end.

Measures: Both aims are read from RMSE relative to the path endpoint on the same draws. An interior optimum occurs when the lowest RMSE along the λ grid sits more than five per cent below the endpoint's. This is a reporting threshold, about two Monte Carlo standard errors at 2500 replications. A selector win occurs when a selector's RMSE sits below the endpoint's by more than three times their combined Monte Carlo standard error. Combining the two as if independent overstates the error of the difference, since the estimates share draws, so the rule errs towards calling a tie. Both are reported as the share of cells in which they occur, and a win carries the selector's RMSE as a ratio to the endpoint's.

Robustness check: Both rules are also applied, with `balnet` alone, to the design of Wong and Chan (2018) as run by Wang and Zubizarreta (2020), matched to our misspecified grid by overlap and confounding strength. The two designs give the same verdict on the selector win in every one of fifteen matched cells and on the interior optimum in all but one (Appendix H).

4 Study 1 Results

4.1 Sample size

We begin with the sample-size grid, under a correctly specified and a misspecified propensity score (Table 4). Balancing weights have lower RMSE than MLE weights in every cell under either specification. At exact balance, ABW collapses to the unaugmented `balnet`, and together they average 68.6% lower RMSE than the Hájek estimator and 33.8% lower than AIPW. Pooled across specifications, $\texttt{balnet}_{10^{-4}}$ no longer beats the oracle in every cell, though it still does under quadratic outcomes (Appendix Table 5). Against OLS, the approximation error of its logistic weights pulls in both directions and cancels out on average under a correctly specified score.

Table 4: $\bar{R}(\Theta)$ and $\bar{s}_m(\Theta)$ of A against B, n = 500 to 50k.

		$\bar{R}(\Theta)$				$\bar{s}_m(\Theta)$	
Estimator (A)	Comparator (B)	All	Linear	Quadratic	Exponential	A	B
Both specifications							
ABW-balnet	balnet	0.1	0.1	0.2	0.1	0.39	0.39
	AIPW-glmnet	33.8	34.8	48.9	12.7	0.39	0.26
	glmnet (Hájek)	68.6	61.6	75.9	66.6	0.39	0.60
balnet	AIPW-glmnet	33.7	34.8	48.9	12.6	0.39	0.26
	glmnet (Hájek)	68.6	61.6	75.8	66.5	0.39	0.60
AIPW-glmnet	glmnet (Hájek)	52.6	41.1	52.7	61.7	0.26	0.60
Correct specification							
ABW-balnet	balnet	0.1	0.0	0.1	0.1	0.00	0.00
	AIPW-glmnet	4.0	-0.2	9.2	2.7	0.00	0.09
	glmnet (Hájek)	69.6	75.3	62.9	69.3	0.00	0.88
balnet	AIPW-glmnet	3.9	-0.2	9.1	2.7	0.00	0.09
	glmnet (Hájek)	69.5	75.2	62.9	69.3	0.00	0.88
AIPW-glmnet	glmnet (Hájek)	68.3	75.3	59.1	68.4	0.09	0.88
Misspecified propensity score							
ABW-balnet	balnet	0.2	0.2	0.3	0.2	0.77	0.78
	AIPW-glmnet	54.3	57.6	71.3	21.7	0.77	0.43
	glmnet (Hájek)	67.6	40.5	84.3	63.6	0.77	0.32
balnet	AIPW-glmnet	54.2	57.5	71.2	21.6	0.78	0.43
	glmnet (Hájek)	67.5	40.4	84.3	63.5	0.78	0.32
AIPW-glmnet	glmnet (Hájek)	29.1	-40.4	45.3	53.5	0.43	0.32

Largest MCSE: 1.9 points ($\bar{R}(\Theta)$), 0.014 ($\bar{s}_m(\Theta)$). The averages are over 30 cells, 15 per specification, at 10 000 Monte Carlo replications.

Under a correctly specified propensity score, the balancing estimators have lower RMSE than every modelling estimator across outcome functions, the one exception a near-zero difference against AIPW under linear outcomes. Here, the difference between modelling and balancing is bias driven while balancing's advantage over the oracle comes through variance, consistent with its theoretical efficiency (Wager 2024). Beyond these sample sizes, the correctly specified grid to $n = 400{,}000$ shows the same ordering, with balancing's lead over modelling widening slightly with n (Appendix Table 1). AIPW also recovers much of the RMSE losses of the unaugmented modelling weights, less so where the shared outcome model is wrong, under the quadratic and exponential outcomes.

Under a misspecified propensity score, both models are fitted on $\tilde{X}$ and are doubly wrong (Table 1). Both balancing estimators keep substantially lower RMSE in every cell, consistent with the findings of Tan (2020b). Their average lead across n runs from 96.9% over the HT estimator at the extreme to 54.3% over AIPW in modelling's best case (Appendix Table 4; Table 4). The lead of

the unaugmented balancing weights over the Hájek estimator is about the same as under a correct score, but its source changes. The unaugmented `balnet` now carries a higher share of bias while the Hájek estimator's error is mostly variance.

Averaging over outcome functions also hides a reversal. Under a correctly specified score, balancing's lead was largest under the linear outcome. Misspecification reverses that ordering. Here the linear outcome is balancing's worst case, misspecification multiplying the RMSE of $\texttt{balnet}_{10^{-4}}$ relative to the Hájek estimator's by 2.4 and pushing it behind the oracle (Table 4; Appendix Table 5). Conversely, it now performs best under the quadratic, where previously its lead over the Hájek estimator was smallest, misspecification dividing the same ratio by 2.4. Following Ben-Michael et al. (2021), the relative loss of $\texttt{balnet}_{10^{-4}}$ here comes from balancing being unable to improve past exact balance on $\tilde{X}$. The Hájek estimator may be spared because its errors sit where the score is small, a region the linear outcome keeps moderate (Tan 2020b).

For the augmented estimators we take the two cases in turn, the outcome model refitted on the true covariates X and both models fitted on $\tilde{X}$. The first is a benchmark an analyst who sees only $\tilde{X}$ cannot reach, and shows the double robustness the augmented estimators provide when the outcome model is correct in both function and covariates.

Figure 1 follows the two augmented estimators with the outcome model refitted on X from $n = 250$ to 10^6 under the linear outcome function, with $\texttt{balnet}_{10^{-4}}$ and the Hájek estimator alongside for reference. The correct outcome model leaves both augmented estimators approximately unbiased beyond the smallest samples, but their RMSE parts as n grows. ABW's falls with n while AIPW's turns upward alongside the Hájek estimator's, whose weights it shares. The gap is the cost of the wrong weights, which a correct outcome model strips of bias but not variance (Ben-Michael et al. 2021). Across n this leaves ABW 52% below AIPW under the linear outcome and about three quarters lower when the outcome function is also misspecified (Appendix Table 4).

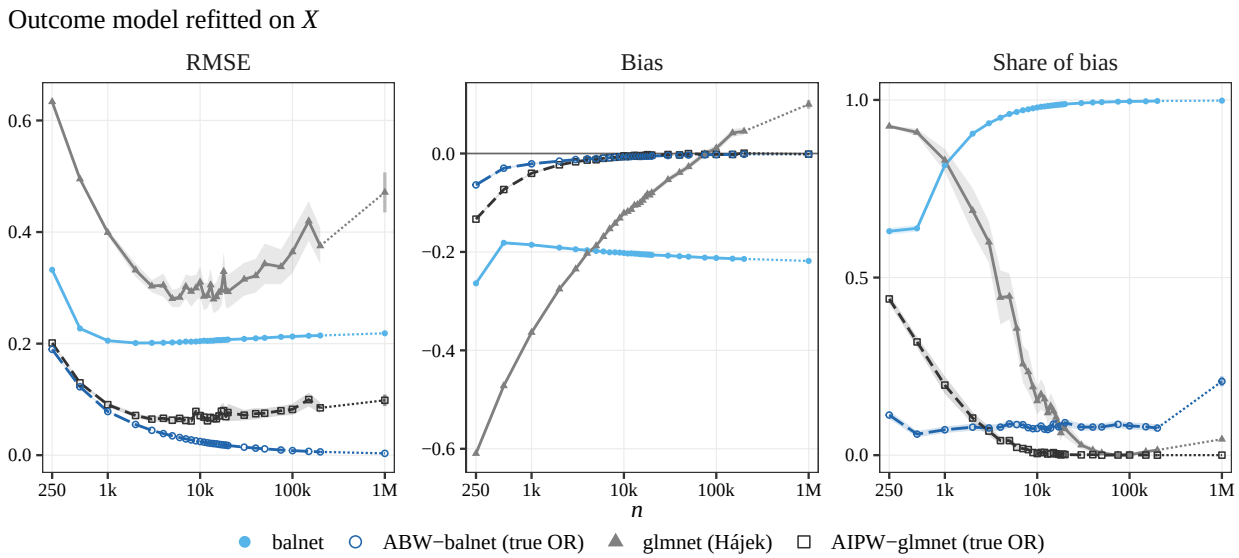


Figure 1: Outcome model refitted on X by n, misspecified propensity score, linear outcome. $n =$ 1M (dotted) is a separate run.

Once both models are fitted on the observed $\tilde{X}$, the double fragility of AIPW is clear. The outcome model that protected it in the refit is now a second source of error. Fitted on $\tilde{X}$, it leaves the

part of the outcome missed by its linear covariates in its residuals, which are then weighted by the misspecified MLE weights, so the bias is the imbalance those weights leave in the residual (Ben-Michael et al. 2021).

Figure 2 follows $\texttt{balnet}_{10^{-4}}$ and the modelling estimators across our three doubly wrong settings. Our balancing estimator converges, but to the wrong value, as is expected of a misspecified estimator. Its variance shrinks towards zero and its bias settles near -0.22, -0.07 and -0.14 under the linear, quadratic and exponential outcome functions (Appendix Table 3). We focus on the Hájek and AIPW estimators, as the poor performance of the HT estimator is well documented (Kang and Schafer 2007; Busso et al. 2014), and leave its results to the appendix.

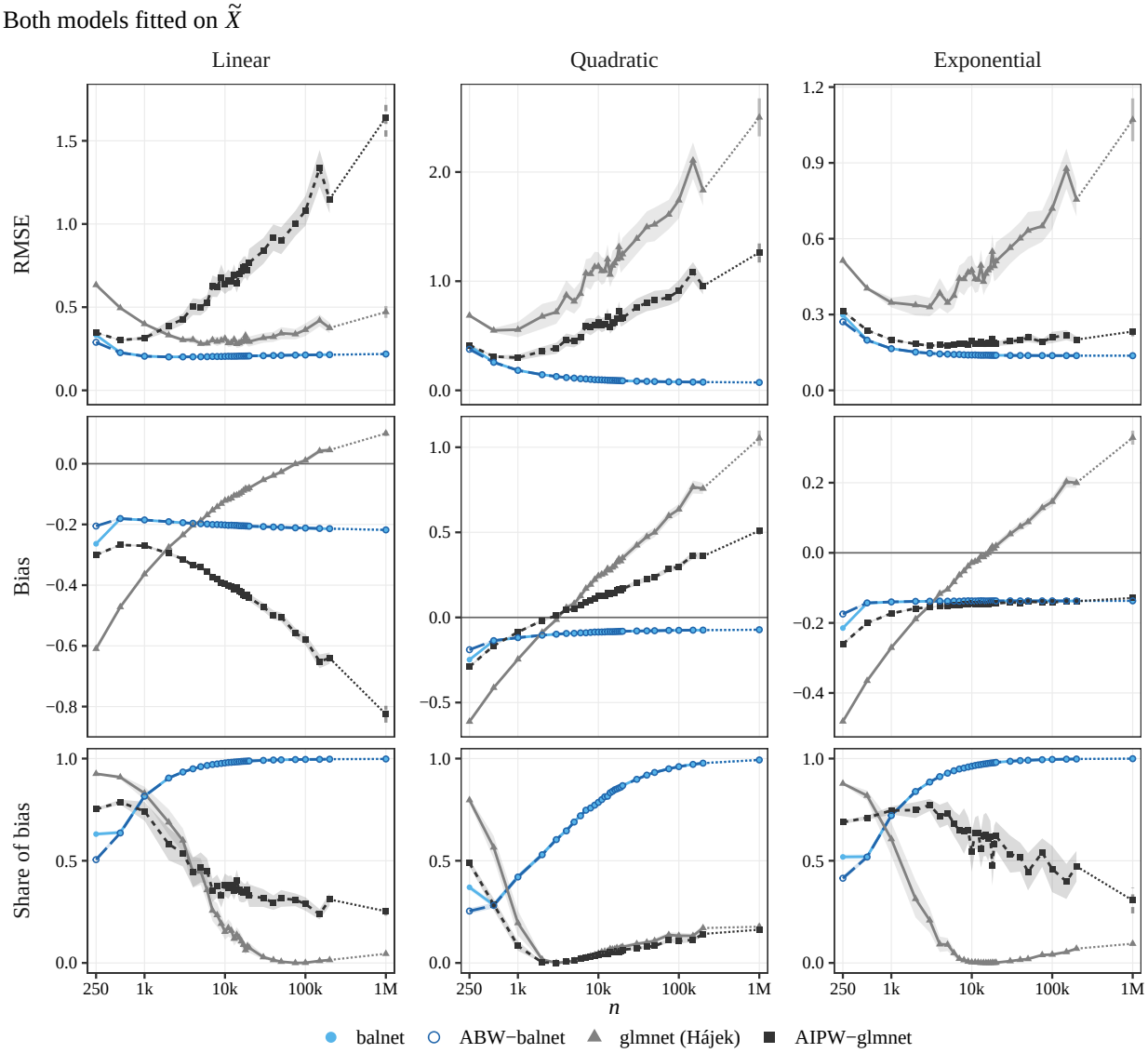


Figure 2: Both models fitted on $\tilde{X}$ by n, misspecified propensity score. n = 1M (dotted) is a separate run. ABW coincides with balnet from $n = 500$.

The Hájek estimator does not settle. Under every outcome function its bias shrinks through zero and then grows positive, and its RMSE passes a minimum and rises to the largest sample. More data makes it worse, a direction Testa et al. (2026) also saw to $n = 1000$. Kang and Schafer (2007) traced the worsening to a few enormous weights where the fitted model misses in the region

of near-zero scores. As n grows those tails lengthen rather than settling, and the share of the sample carrying the weight shrinks with it. At $n = 500$ the MLE weights are no more extreme than the true score's, with a similar largest weight and a larger effective sample size (Appendix Table 2), so the tails cannot explain the bias. Here, the bias comes from the imbalance the weights leave in $\tilde{X}$.

Under the linear outcome function the balancing weights settle at the same bias alone and augmented, near -0.2, well below the Hájek estimator. Augmented, they reduce to OLS on $\tilde{X}$, as exact balance makes the outcome model a linear adjustment on $\tilde{X}$ (Bruns-Smith et al. 2026). Unaugmented, they differ from that OLS only by the approximation error, so both carry the part of the outcome not linear in $\tilde{X}$. The double fragility of AIPW is pronounced here. Its path starts ahead of the Hájek estimator, slips behind it by $n = 2000$, and loses ground from there to over three times its RMSE by $n = 10^6$ (Appendix Table 3). This is the ordering Kang and Schafer (2007) report up to $n = 1000$, here holding to 10^6. For AIPW, two wrong models have previously been found to beat one when the outcome model's misspecification is moderate, though every bias still scales with the propensity score error (Waernbaum and Pazzagli 2023). Here the two errors compound instead.

Under the quadratic and exponential outcome functions the outcome model is also wrong in form, yet AIPW now beats the Hájek estimator, likely because the linear model in $\tilde{X}$ picks up part of the outcome's curvature, so its residual is less imbalanced than the outcome itself (Ben-Michael et al. 2021). It still stays well behind ABW, and $\texttt{balnet}_{10^{-4}}$ pulls further ahead of the Hájek estimator. Under the exponential, AIPW is the one modelling series that settles, at a bias near -0.13. The outcome's $\exp(X_1/2)$ term is linear in $\tilde{X}_1$, so the outcome model may fit exactly the part of the outcome where most of the score's signal sits. Refitting the outcome model on X does not rescue AIPW under either outcome function, and under the quadratic it makes both augmented estimators worse at large n. The same wrong-form model costs almost nothing under a correctly specified propensity score, where both augmented estimators are nearly unbiased, so what AIPW appears unable to carry is the pairing of two wrong models.

Misspecification costs the balancing weights a fixed bias that more data does not shrink. For modelling, it costs the weights a bias that falls, crosses zero and grows again while their tails lengthen with n, and that bias compounds in AIPW under the linear outcome function. Balancing stays ahead at every n under every outcome function, alone and augmented, and the bias of `cv.glmnet` has not settled by $n = 10^6$.

4.2 Overlap

Pooled over overlap and both specifications, balancing estimators have lower RMSE on average, and $\texttt{balnet}_{10^{-4}}$ has lower RMSE than the Hájek estimator in every cell (Table 5). ABW has on average 20.2% lower RMSE than $\texttt{balnet}_{10^{-4}}$ and 32.5% lower than AIPW, while the unaugmented `balnet` has on average 23.1% lower than AIPW and 59.9% lower than the Hájek estimator.

AIPW has lower RMSE than $\texttt{balnet}_{10^{-4}}$ on average only where common support is practically violated and the propensity score is correctly specified. There MLE's performance is also unobservable at $c = 0.05$, where 81.8% of the sample have a propensity score below 0.01 or above 0.99 (Appendix Table 6).

Where we assume common support holds, both balancing estimators have lower RMSE on av-

Table 5: $\bar{R}(\Theta)$ and $\bar{s}_m(\Theta)$ of A against B, $c = 0.05$ to 2, $n = 1000$ and 10 000, both specifications.

		$\bar{R}(\Theta)$				$\bar{s}_m(\Theta)$	
Estimator (A)	Comparator (B)	All	Linear	Quadratic	Exponential	$m = A$	$m = B$
balnet	AIPW-glmnet	23.1	22.0	35.5	9.6	0.55	0.34
	glmnet (Hájek)	59.9	57.2	64.7	57.5	0.55	0.67
ABW-balnet	balnet	20.2	21.6	20.1	19.0	0.44	0.58
	AIPW-glmnet	32.5	32.4	43.4	19.6	0.46	0.34

Largest MCSE: 0.4 points ($\bar{R}(\Theta)$), 0.004 ($\bar{s}_m(\Theta)$). The averages are over 84 cells, 42 per specification, at 10 000 Monte Carlo replications. Pairs with glmnet omit the six correct-specification cells at $c = 0.05$, where glmnet lacks more than 1% of its estimates.

erage than the modelling estimators, with ABW averaging 7.5% lower RMSE than AIPW and 0.2% lower than $\texttt{balnet}_{10^{-4}}$ across outcome functions (Table 6). Balancing's advantage here holds across nonlinear outcomes, though for the linear outcome function, where the shared outcome model is correct, ABW has 0.6% higher RMSE than AIPW while $\texttt{balnet}_{10^{-4}}$ has 0.8% higher. For misspecified propensity scores, both balancing estimators have RMSE around 51% lower than AIPW, as on the sample-size grid.

Table 6: $\bar{R}(\Theta)$ and $\bar{s}_m(\Theta)$ of A against B where common support holds.

		$\bar{R}(\Theta)$				$\bar{s}_m(\Theta)$	
Estimator (A)	Comparator (B)	All	Linear	Quadratic	Exponential	A	B
Correct specification							
balnet	AIPW-glmnet	7.2	-0.8	15.8	6.0	0.02	0.08
ABW-balnet	balnet	0.2	0.2	0.3	0.2	0.02	0.02
	AIPW-glmnet	7.5	-0.6	16.1	6.2	0.02	0.08
Misspecified propensity score							
balnet	AIPW-glmnet	51.3	53.8	65.9	26.8	0.76	0.46
ABW-balnet	AIPW-glmnet	51.4	53.9	66.0	27.0	0.76	0.46

Largest MCSE: 0.7 points ($\bar{R}(\Theta)$), 0.008 ($\bar{s}_m(\Theta)$). Overlap $c = 0.5$ to 2 at $n = 1000$ and 10 000. The averages are over 24 cells per specification, at 10 000 Monte Carlo replications.

Where we assume common support is violated, under a correctly specified score ABW has 21.7% lower RMSE than AIPW while the unaugmented `balnet` has around 1.5 times the RMSE of AIPW (Table 7). This ordering holds under every outcome function, with the gap to $\texttt{balnet}_{10^{-4}}$ widest under the linear outcome. Under a misspecified score, for all $c \leq 0.25$ ABW has 37.7% lower RMSE than AIPW on average, while $\texttt{balnet}_{10^{-4}}$ has 29.3% lower. As in simulations with practical or outright positivity violations, $\texttt{balnet}_{10^{-4}}$ has lower RMSE than the Hájek estimator (Chattopadhyay et al. 2020; Keele et al. 2026) but is biased where its path truncates. Under a misspecified score the lowest RMSE at the weakest overlap does not improve with n (Appendix Table 8).

As overlap degrades below $c = 0.5$, the share of bias rises for both balancing estimators. Under correct specification, averaged over both sample sizes, the unaugmented `balnet` jumps from a 1% share at $c = 0.5$ to 41% at $c = 0.25$ under linear outcomes (Figure 3). This average hides a split by sample size, with the share 83% at $n = 1000$, where every path stops short, and zero at $n = 10{,}000$, where nearly every path reaches its target (Appendix Table 7). In contrast, ABW moves from unbiased to a 6% share, rising to 28% by $c = 0.05$ where $\texttt{balnet}_{10^{-4}}$ is at 97%. This pattern holds across outcomes, except that ABW under quadratic outcomes becomes more variance driven on either side of $c = 0.25$.

Table 7: $\bar{R}(\Theta)$ and $\bar{s}_m(\Theta)$ of A against B where common support is violated.

		$\bar{R}(\Theta)$				$\bar{s}_m(\Theta)$	
Estimator (A)	Comparator (B)	All	Linear	Quadratic	Exponential	A	B
Correct specification							
balnet	AIPW-glmnet	-49.6	-77.5	-27.6	-47.8	0.65	0.18
ABW-balnet	AIPW-glmnet	21.7	12.7	30.5	20.9	0.13	0.18
Misspecified propensity score							
balnet	AIPW-glmnet	29.3	35.6	33.0	17.9	0.90	0.62
ABW-balnet	AIPW-glmnet	37.7	44.2	42.3	24.8	0.87	0.62

Largest MCSE: 2.0 points ($\bar{R}(\Theta)$), 0.004 ($\bar{s}_m(\Theta)$). Overlap c = 0.05 to 0.25 at n = 1000 and 10 000. The averages are over 12 cells under correct specification and 18 under misspecification, at 10 000 Monte Carlo replications.

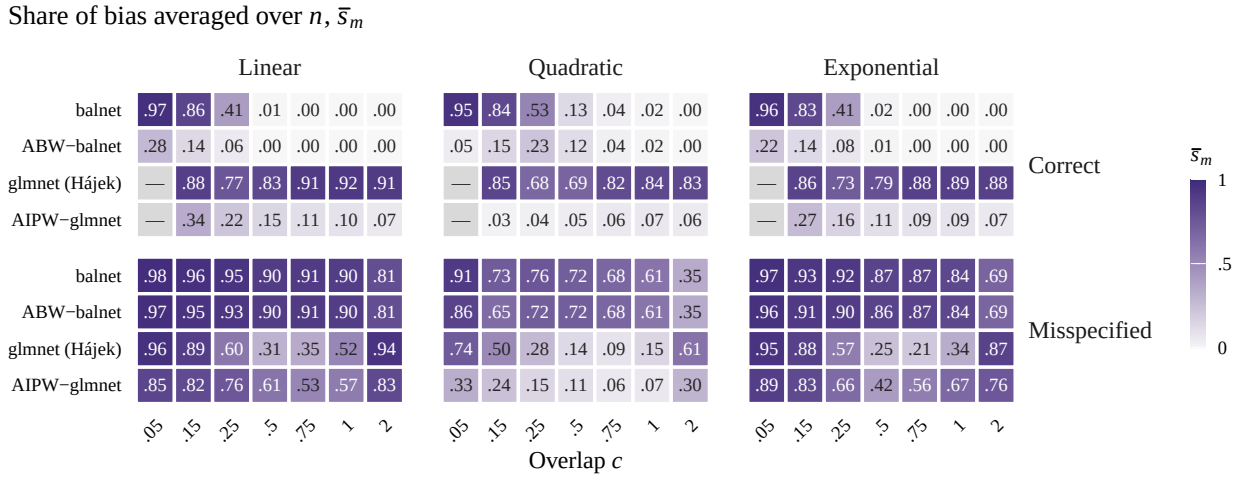


Figure 3: Share of bias by c, averaged over n = 1000 and 10 000, both specifications. glmnet (HT) not drawn. Grey cells are left out for missing estimates.

Appendix Figure 3 follows the estimators as overlap improves in small and large samples. Across all settings, ABW has lower RMSE than the unaugmented `balnet` wherever the path of `balnet` stops short of exact balance, increasingly so under a correctly specified score as overlap degrades, and with more data the path reaches exact balance at poorer overlap.

Where we assume common support holds, ABW has the lowest RMSE of the modelling and balancing estimators in small samples under every outcome. In these samples AIPW also has a lower RMSE than $\texttt{balnet}_{10^{-4}}$ under a correctly specified score, and this extends to larger samples as overlap degrades. Under a misspecified propensity score the advantage of $\texttt{balnet}_{10^{-4}}$ over AIPW extends to poorer overlap, with a lower RMSE at $n = 5000$ from $c = 0.1$, even where its path truncates, and at every c under linear outcomes.

Where we assume common support is practically violated, $\texttt{balnet}_{10^{-4}}$ truncates at every n, with a higher RMSE than ABW in both samples, far more so under a correctly specified score. Under a misspecified score AIPW has a lower RMSE than ABW at the weakest overlap under the quadratic outcome and the two are level under the exponential, but as overlap improves ABW takes over, earlier in the larger sample, consistent with approximate residual balancing outperforming AIPW in the misspecified setting of Athey et al. (2018), which varied dimension. At the weakest overlap no weighting estimator removes half the confounding in either sample, as in Keele et al. (2026),

where no method overcame outcome misspecification once overlap was poor.[6]

4.3 High-dimensional settings

Table 8: $\bar{R}(\Theta)$ of A against B and $\bar{s}_m(\Theta)$ of A over all p/n, at exact balance and once the path stops short, $n = 500$ and 1000.

		$\bar{R}(\Theta)$			$\bar{s}_m(\Theta)$, $m = A$	
Estimator (A)	Comparator (B)	All	Exact balance	Path truncates	Exact balance	Path truncates
Correct specification						
balnet	glmnet (Hájek)	49.1	66.9	36.9	0.01	0.89
	AIPW-glmnet	-48.0	3.0	-82.8	0.01	0.89
ABW-balnet	balnet	41.6	0.0	55.4	0.01	0.46
	AIPW-glmnet	13.6	3.0	18.4	0.01	0.46
Misspecified propensity score						
balnet	glmnet (Hájek)	39.2	55.5	28.9	0.57	0.89
	AIPW-glmnet	-7.9	25.4	-29.8	0.57	0.89
ABW-balnet	balnet	23.8	0.2	33.5	0.57	0.75
	AIPW-glmnet	17.8	25.5	13.7	0.57	0.75

Cells per specification: 36 in all, 12 while exact balance holds and 24 once the path truncates, at 10 000 Monte Carlo replications. Largest MCSE: 0.8 points ($\bar{R}(\Theta)$), 0.019 ($\bar{s}_m(\Theta)$).

As covariates are added, balancing keeps its advantage over modelling on average, alone and augmented (Table 8). $\texttt{balnet}_{10^{-4}}$ has lower RMSE than `cv.glmnet` (Hájek) in every cell, by 49.1% on average under a correctly specified score and 39.2% under misspecification. Augmentation narrows this to 13.6% and 17.8% for ABW over AIPW, and on average AIPW overtakes the unaugmented `balnet`.

These averages hide the expected change at $p/n = 0.5$, from which every $\texttt{balnet}_{10^{-4}}$ path stops short of exact balance. Bias then makes up most of its error, with its share rising from 0.01 to 0.89 under a correctly specified score (Table 8). Here, its lead over the Hájek estimator nearly halves, from 66.9% to 36.9%, and it also slips behind the oracle (Appendix Table 11).

ABW again collapses to $\texttt{balnet}_{10^{-4}}$ at exact balance. Once the path stops short, the outcome model adjusts for the remaining imbalance, and ABW's RMSE is 55.4% and 33.5% lower than that of $\texttt{balnet}_{10^{-4}}$. Unlike $\texttt{balnet}_{10^{-4}}$, ABW stays ahead of AIPW, by 18.4% and 13.7%. Its lead widens from $p/n = 0.5$ under a correctly specified score but narrows under misspecification.

Once the path stops short, the lead of $\texttt{balnet}_{10^{-4}}$ over the Hájek estimator narrows as p grows under all settings (Figure 4). Both estimators grow more biased, but at different rates. As expected, the bias of the Hájek estimator grows gradually, while that of $\texttt{balnet}_{10^{-4}}$ grows faster as its leftover imbalance increases with p (Zhao 2019). Its nonzero coefficients stop growing with p once the path stops short (Appendix Table 13; Appendix Table 14). Our grid extends the comparison of Tan (2020b) to higher dimensions, up to $p/n = 4$, where the narrowing we find between the unaugmented estimators mirrors the result of Athey et al. (2018) for approximately balancing weights.

Augmentation does not remove the whole cost of the leftover imbalance, with ABW's bias still

[6] At $c = 0.01$ the unadjusted difference is -2.06 (linear, quadratic) and -1.56 (exponential) at both n, against a lowest weighting RMSE of 1.08 (design values, weighting RMSEs in Appendix Table 9).

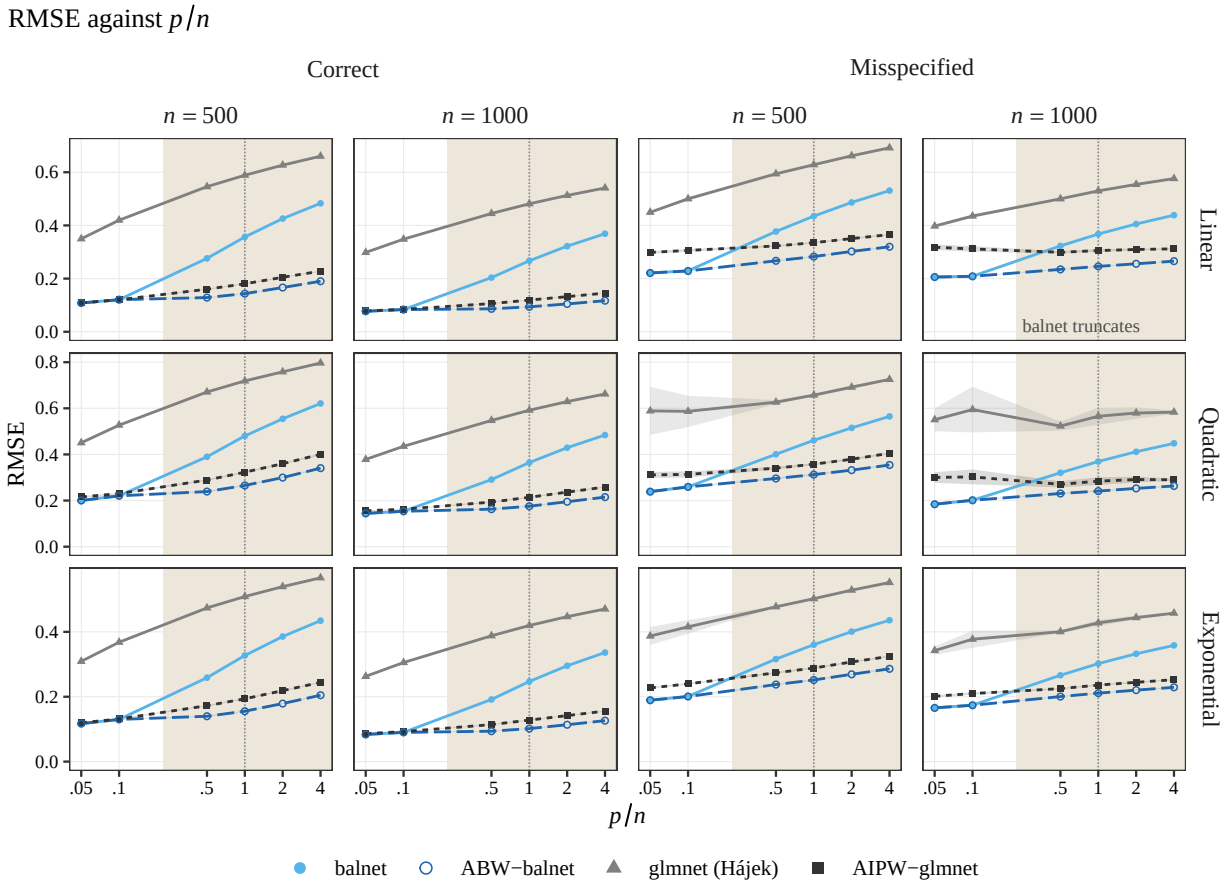


Figure 4: RMSE against p/n at $n = 500$ and 1000, both specifications. Shading marks where the balnet path stops short, the dotted line $p = n$.

growing with p from $p/n = 0.5$ (Appendix Figure 4). Its lead over AIPW holds steady at 15% to 21% under a correctly specified score (Appendix Table 11), but narrows as p grows under misspecification, most clearly for the linear outcome. The narrowing comes from both sides, as ABW's bias grows faster than AIPW's once the path stops short, while AIPW's variance falls with p and offsets its own growing bias, most of all at $n = 1000$. To our knowledge this narrowing has not been reported, and the augmented balancing weights of Athey et al. (2018), which target balance without a propensity score, show a steady lead over AIPW.[7]

The cost of misspecification to the MLE estimators shrinks as covariates are added. For the Hájek estimator this cost falls in every series, and has vanished or reversed for nonlinear outcomes by $p/n = 4$. Though the relative cost falls, Hájek's RMSE still generally rises with p, consistent with the findings of Athey et al. (2018). As discussed earlier, misspecification costs AIPW more than the Hájek estimator at low p, and this cost too shrinks as p grows (Figure 4).

With a denser covariate profile: With four confounders, $\nu = 1$ and independent covariates, the sparsity grid shares the propensity score of the p/n grid. From $p/n = 0.5$ every `balnet` path stops short of exact balance, whatever s, ν or ρ, so exact balance and the truncated path are reported apart throughout (Figure 5). `balnet` again has lower RMSE than the Hájek estimator in every cell under both scores, and by more than three MCSE in all but one under Tan's transformation, whatever the

[7] Unlike ours, their AIPW used elastic-net propensity scores trimmed at 0.05 and 0.95 (Athey et al. 2018).

number of confounders, their decay or the covariate correlation.[8] ABW keeps a lead over AIPW on average under independent and AR(1)-correlated covariates, but under equicorrelation the two stay within a few per cent.

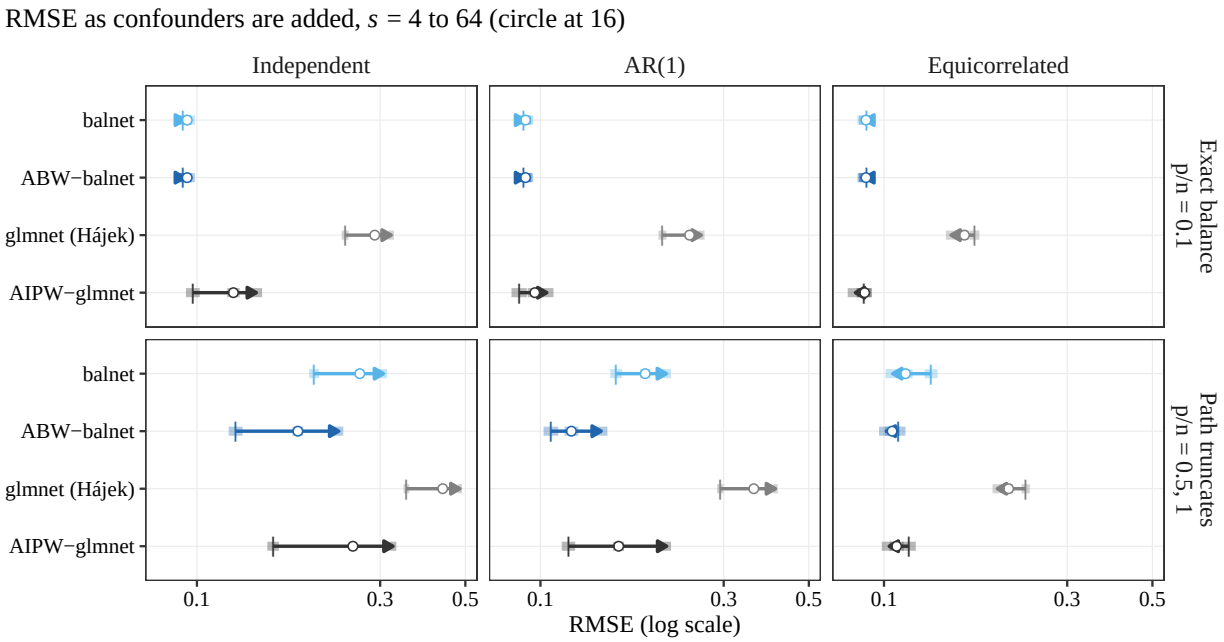


Figure 5: RMSE by s and covariance structure, correct specification, at exact balance and once the path stops short, pooled over outcomes and decay rates. glmnet (HT) not drawn.

Dynamics across s: While exact balance holds, `balnet` stays close to unbiased in every cell, with absolute bias at most 0.006 for the linear outcome and 0.028 for the quadratic, at every s, ν and ρ (Appendix Table 15). Here, the Hájek estimator grows more biased as confounders are added under independent and AR(1)-correlated covariates, so the lead of `balnet` widens with s, from 62.2% to 71.1% under independent and from 56.5% to 65.3% under AR(1)-correlated covariates at $n = 1000$. The lead is larger under slower decay, whereas under equicorrelation it narrows instead (Appendix Table 17). As expected, the error of the MLE weights grows with the number of active covariates (Tan 2020a). Exact balance instead removes the bias for the linear outcome at any s.

Once the path stops short, all four estimators are biased downwards (Appendix Table 15). The bias of `balnet` grows as confounders are added and, beyond four, as their decay slows, and is smaller under AR(1)-correlated covariates, consistent with the findings of Zhao (2019) for the number of confounders and of Athey et al. (2018) for correlation. Under equicorrelation it falls with s instead. `balnet` still keeps a lead of 37% to 48% over the Hájek estimator at every s, though its error turns to bias, with a share rising from 0.73 to 0.86 under independent and from 0.60 to 0.76 under AR(1)-correlated covariates (Appendix Table 18). Its linear bias is nearly the full imbalance bound with four confounders but between a half and three quarters of it with 64 under independent and AR(1)-correlated covariates (Appendix Table 16).

[8] In the remaining cell ($n = 1000$, $s = 4$, $\nu = 1$, AR(1), quadratic outcome) the RMSE of `balnet` is 40% of the Hájek estimator's, but heavy-tailed Hájek estimates leave the ratio within three MCSE of one, so it is not counted as a win.

The gain from augmentation follows the same split. Once the path stops short, ABW has lower RMSE than AIPW in every cell under independent and AR(1)-correlated covariates, and its lead grows with s, from 20.1% to 28% by 16 confounders under independent and from 10.0% to 32.5% under AR(1)-correlated covariates (Appendix Table 18). Under these two structures both augmented estimators grow more biased as confounders are added, AIPW faster. As expected, AIPW loses ground as the propensity model it relies on grows denser (Athey et al. 2018). This is consistent with the findings of Bradic et al. (2019), where the error of AIPW rose with the number of active propensity covariates while that of augmented balancing weights did not. While exact balance holds, the same pattern appears at $n = 1000$, from a lead of 5.8% under independent and none under AR(1)-correlated covariates at four confounders to 35.0% and 12.5% at 64.

The unaugmented `balnet` also catches AIPW as confounders are added. Once the path stops short, `balnet` has 1.3 times the RMSE of AIPW at four confounders under independent and AR(1)-correlated covariates. At 64 it is level with AIPW under AR(1)-correlated covariates and ahead under independent ones, and the gap shrinks with s under every covariance structure (Appendix Table 18). Here, the bias of AIPW grows with s faster than that of `balnet`, while under equicorrelation the bias of `balnet` falls faster.

Tan's transformation costs `balnet`, not the Hájek estimator, and only while exact balance holds. There the RMSE of `balnet` rises by about a half under independent covariates, two thirds for the linear outcome, and by a fifth under AR(1), so the lead narrows from 63.6% to 49.6% under independent covariates and little elsewhere (Appendix Table 19). Here, exact balance on the transformed covariates does not guarantee balance in the original ones, in which the outcome is linear, so part of the confounding bias can return (Zhao 2019). This differs from the findings of Tan (2020b), where balancing weights had similar or smaller MSE under a correct score and a substantial advantage under a Kang–Schafer misspecification. Under the transformation the lead of ABW over AIPW also narrows with s at exact balance, the reverse of its direction under a correct score. The direction matches the findings of Tan (2020a) under a misspecified score, though their outcome model was correct.

Equicorrelated covariates break the pattern. While exact balance holds, the Hájek estimator grows less biased as confounders are added, so the lead of `balnet` narrows from 47.7% to 40.4% instead of widening, and once the path stops short the bias of `balnet` falls with s too (Appendix Table 17; Appendix Table 18). ABW and AIPW stay within a few per cent of each other at exact balance and once the path stops short, and Tan's transformation costs `balnet` little. We read this through Z_0, the common factor every equicorrelated covariate shares, which makes up about three quarters of the variance of the propensity index at four confounders and over nine tenths at 64 (Appendix Table 20). Here, a sparse fit already captures most of the confounding, so balancing more columns adds little.

4.4 Outcome signal

Under a correctly specified propensity score, the ranking of $\texttt{balnet}_{10^{-4}}$ and the Hájek estimator reverses as the outcome signal grows against the fixed noise (Appendix Table 21). As only the ratio of signal to noise decides the ranking, a low κ reads as a noisy outcome. In the noisiest cells, $\kappa \leq 0.05$, $\texttt{balnet}_{10^{-4}}$ has 9% to 12% higher RMSE, and once the noise falls back, from $\kappa = 0.2$,

it has lower RMSE, by 62% to 74% at $\kappa = 1$. Here, the bias of the Hájek estimator grows with the signal while $\texttt{balnet}_{10^{-4}}$ stays close to unbiased, but exact balance leaves it a smaller effective sample size (median 345 against 430 per arm, Appendix Table 22), which decides the ranking while the noise dominates. Its 32.9% advantage pooled over κ (Table 9) averages over the cells where its RMSE is higher. The oracle reverses against $\texttt{balnet}_{10^{-4}}$ in the same way, so the efficiency advantage of exact balance holds only once the signal clears the noise.

Table 9: $\bar{R}(\Theta)$ and $\bar{s}_m(\Theta)$ of balnet against glmnet (Hájek), correct specification, $\kappa = 0.02$ to 1.

		$\bar{R}(\Theta)$				$\bar{s}_m(\Theta)$	
Estimator (A)	Comparator (B)	All	Linear	Quadratic	Exponential	$m = A$	$m = B$
balnet	glmnet (Hájek)	32.9	35.9	32.5	30.3	0.00	0.47

Largest MCSE: 0.1 points ($\bar{R}(\Theta)$), 0.002 ($\bar{s}_m(\Theta)$). The averages are over 24 cells at 10 000 Monte Carlo replications.

Under a misspecified propensity score both estimators are biased in proportion to the signal, as the bias acts through the outcome (Zhao 2019; Waernbaum and Pazzagli 2023), but the Hájek estimator is about twice as biased. $\texttt{balnet}_{10^{-4}}$ has lower RMSE at every noise level but the highest, by 49% to 68% at $\kappa = 1$, and the two are level at $\kappa = 0.02$. There the Hájek estimator has the lower median absolute error, as a few extreme weights lift its RMSE.

Augmentation again leaves $\texttt{balnet}_{10^{-4}}$ unchanged, with the RMSE of ABW within 0.001 of it in every cell. Under a correctly specified score the gain for the Hájek estimator grows as the noise falls, from none at $\kappa = 0.02$, where the outcome model often keeps no covariate, to 59% to 74% lower RMSE at $\kappa = 1$. The augmented pair reverses as the unaugmented one does, with ABW ahead of AIPW only from $\kappa = 0.2$, by under 8%. Under misspecification neither pair reverses, and ABW stays ahead of AIPW at every noise level but the highest, by 18% to 43% at $\kappa = 1$.

In a noisy outcome, then, the balance at the end of the `balnet` path costs more in variance than it removes in bias, as anticipated where the best amount of balance depends on the noise relative to the signal (Zhao 2019).

4.5 Coverage

Figure 6 shows what the biases of Study 1 cost a confidence interval along n. Under a correctly specified score $\texttt{balnet}_{10^{-4}}$, ABW and the oracle sit at nominal at every n, and AIPW reaches it by $n = 10{,}000$. The Hájek estimator never rises above 0.44, as the shrinkage bias of its cross-validated lasso score does not fall with n while its sampling error does (share of bias 0.88, Table 4). Under misspecification the picture inverts. The balancing estimators collapse to zero by $n = 10{,}000$ for the linear and exponential outcomes and more slowly for the quadratic, as the intervals zero in on their fixed bias. The Hájek and AIPW estimators instead reach nominal from $n = 5000$, AIPW from 10,000 under the exponential outcome. In our design the largest MLE weight grows with n while the true score's does not, from a median under ten at $n = 500$ to several hundred at 50,000 (Appendix Table 2), so the Hájek and AIPW estimates carry heavy tails that inflate EmpSE beyond the spread of a typical estimate.

Along overlap, coverage depends on the balance achieved (Appendix Figure 2). Under a correctly specified score $\texttt{balnet}_{10^{-4}}$ holds nominal coverage where common support holds and loses it

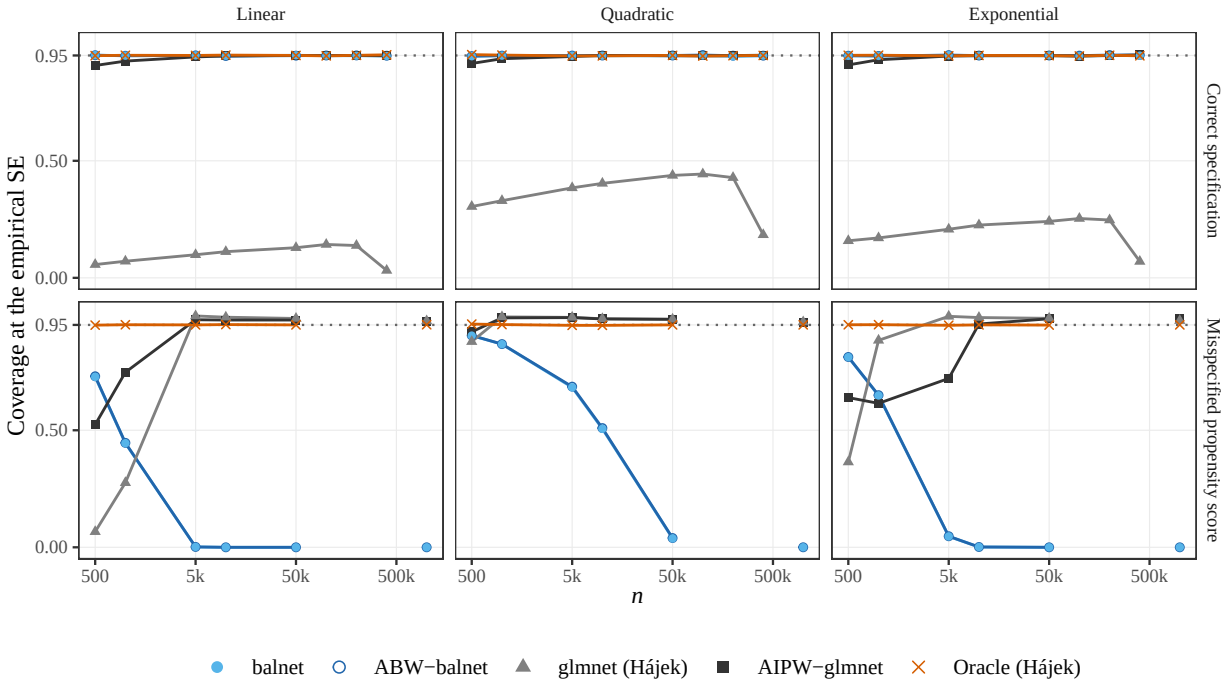


Figure 6: Coverage at the empirical SE by n, both specifications. The dotted line is the nominal 95%, n = 1M (misspecified) a separate run.

where it is practically violated, AIPW loses it only at $c \leq 0.15$ under the linear and exponential outcomes, and ABW stays at or above 0.83 at every c. The Hájek estimator sits below a half at every $c \geq 0.75$, and at $c = 0.05$ even the oracle falls below 0.15. Under misspecification the balancing estimators keep partial coverage at $n = 1000$, rising with overlap from near zero at $c \leq 0.15$ to between 0.7 and 0.9 at $c = 2$, before the collapse with n seen above. The Hájek and AIPW estimators reach nominal only mid-grid and lose it at both ends, under practical violations and at $c = 2$, where the weights are no longer extreme and their bias is exposed.[9]

Noise also obscures bias (Appendix Figure 5). At low κ the noise dominates EmpSE and every estimator reaches nominal coverage. A stronger outcome signal lowers the coverage of a biased estimator, as its bias grows while EmpSE stays fixed. Under a correctly specified score both balancing estimators and AIPW hold near nominal coverage at every κ, while the Hájek estimator falls to between 0.07 and 0.34 at $\kappa = 1$. Under misspecification every estimator is biased and loses coverage as the signal grows, the balancing estimators to 0.45 at $\kappa = 1$ for the linear outcome. The quadratic outcome is the exception, where the Hájek and AIPW estimators stay above nominal. The weights do not depend on the outcome, so the same tails apply, and under the quadratic outcome they give the Hájek estimator an EmpSE three times the linear outcome's while its bias is smaller against it.

Coverage does not change the RMSE orderings above, but illustrates how the estimators' errors would reach a confidence interval in application. For the balancing weights it is a fixed bias no variance estimator can correct, and for the MLE estimators heavy tails can widen the interval until

[9]Under misspecification at $n = 10{,}000$ the largest MLE weight in the treated arm has a median of 15 at $c = 2$, against 98 at $c = 1$ and about 1450 at $c = 0.05$, and the arm's effective sample size is about 4600 of 5000. EmpSE of the Hájek estimator falls to 0.03 while its bias stays near -0.15 for the linear outcome.

it includes the true τ. Where the signal is weak `balnet`'s bias is small against EmpSE, so a larger λ can trade it for variance at little cost to coverage. Study 2 asks where that RMSE-minimising λ lies inside the path, and whether any selector finds it.

5 Study 2 Results

5.1 Overview

Table 10 shows how often the RMSE-minimising λ comes before the path endpoint and how often a selector improves on it. As expected, both become more common at weaker outcome signals, where the endpoint's cost in variance stays fixed by noise while the bias it removes falls with the signal. With no signal both estimators have an interior optimum in every cell, in line with Study 1. From $\kappa = 1/4$ the two diverge, with `balnet`'s share falling from 35% to 8% at full signal while ABW's holds near 60%. On average across our outcome signal levels, an interior optimum occurs in 61% of cells for `balnet` and 79% for ABW.

Table 10: Interior optima and selector wins for balnet and ABW, (A) by grid, (B) by κ.

			Interior optimum (%)		Selector win (%)		Win given an interior optimum (%)	
Grid	Factor varied	Cells	balnet	ABW	balnet	ABW	balnet	ABW
A. By grid								
Overlap	c	18	0	72	0	67	–	92
Outcome noise	κ	21	43	43	38	38	89	89
Noise × overlap	κ, c	84	60	85	57	83	96	99
Penalty form × overlap	α, c	420	58	82	53	80	92	97
Sample size × overlap × penalty	n, c, α	336	58	82	57	81	98	99
Large n × overlap	n, c	84	73	100	58	100	80	100
Dimension × overlap × penalty	p, c, α	336	68	75	65	73	95	97
Misspecification × overlap	c	126	53	61	51	62	96	99
Denser profile	s, covariance	189	66	79	66	78	98	99
Denser profile under Tan's transformation	s, covariance	189	66	83	65	81	94	97
All grids		1803	61	79	58	78	94	98
B. By outcome signal κ, all grids								
κ = 0		255	100	100	100	100	100	100
κ = 1/32		255	99	99	98	98	98	98
κ = 1/16		255	96	96	92	93	96	97
κ = 1/8		255	80	82	80	81	99	99
κ = 1/4		255	35	62	26	60	69	94
κ = 1/2		255	13	58	9	56	59	97
κ = 1		273	8	59	7	60	82	99

Interior optimum: the path minimum is more than 5% below the endpoint's RMSE. Selector win: a selector's RMSE is below the endpoint's by more than three times their combined MCSE; – where a grid has no interior optimum. Cells per estimator: design cells × κ levels × three outcomes; pooled rows weight grids by cells. Without an interior optimum a selector won in 9 of 700 cells for balnet and 5 of 373 for ABW. 2500 replications per cell, 1000 at n = 10 000.

At least one selector improves on the endpoint in 58% of cells for `balnet` and 78% for ABW, almost always where an interior optimum exists. `cv.bloss` wins most often for both estimators, followed by `boot.inf` (Appendix Table 24). Still, none of the five has the lowest RMSE in more than around 30% of cells. Applied naively, however, `boot.inf` appears the stronger candidate for `balnet`, with RMSE on average 5% below the endpoint's over all settings against 1% for `cv.bloss`. For ABW the two are level on average, at 15% and 16%, so `cv.bloss` remains our choice on its win count.

5.2 Outcome signal and overlap

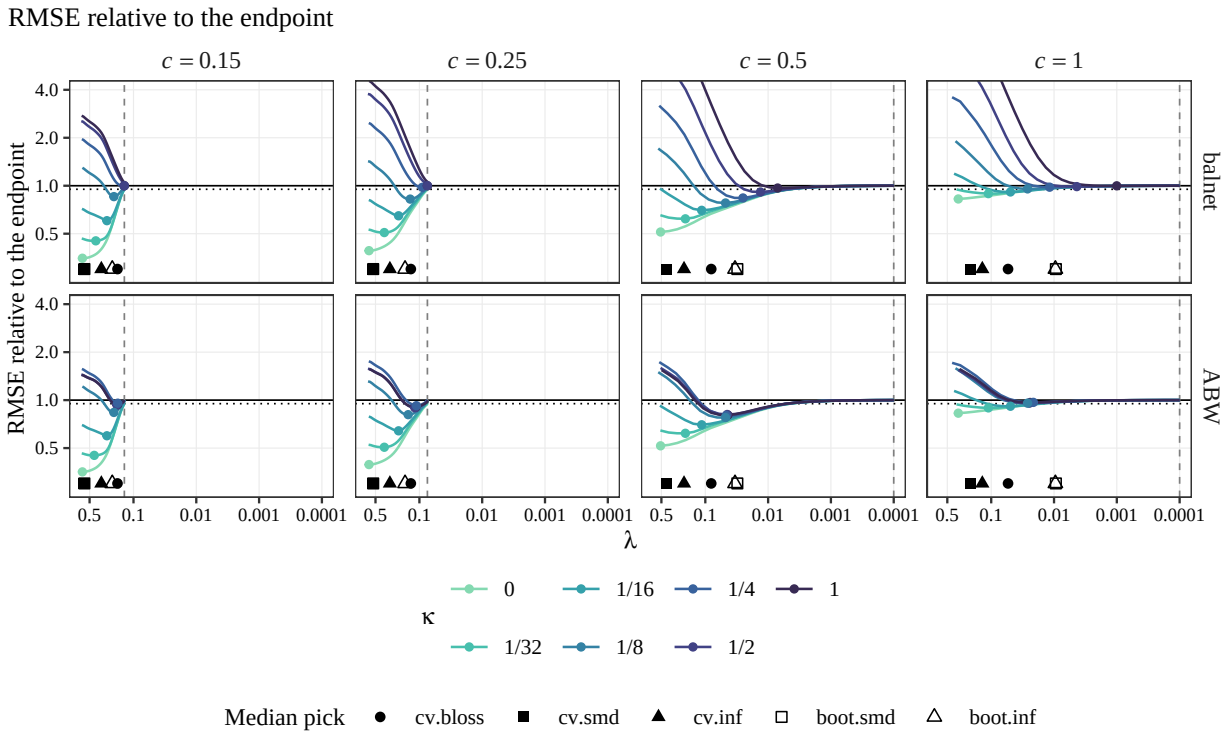


Figure 7: RMSE along the λ path relative to the endpoint, by c and κ, linear outcome, $n = 1000$, $p = 50$. Points mark each minimum, the dotted line the interior rule, the dashed line the median endpoint λ and symbols each selector's median λ. Ratios above 4 are cut off.

Interior optima: Figure 7 traces RMSE along the path relative to the endpoint by overlap and outcome signal. We find that the outcome signal decides whether `balnet` has an interior optimum, while overlap sets how far below the endpoint's RMSE it lies and how strong a signal it survives. With no signal the path minimum lies 65% below the endpoint at $c = 0.15$ against 17% at $c = 1$, the larger gain under limited overlap we anticipated (Wang and Zubizarreta 2020). This gain closes as the signal grows, first at $c = 1$, where interior optima vanish from $\kappa = 1/8$, and everywhere by full signal. Under practical violations of common support, however, interior optima vanish sooner, lasting to $\kappa = 1/2$ at $c = 0.5$ but not at $c \leq 0.25$. There the path already stops short of exact balance, so the endpoint weights are already approximately balanced and a larger λ has little left to gain for `balnet`. This suggests λ selection pays most in between, where the path reaches exact balance but overlap is limited.

For ABW the roles reverse, and overlap rather than signal largely decides whether an interior optimum exists. At $c \leq 0.5$ it has one at every signal level, in all but one cell, while at $c = 1$ its optima vanish from $\kappa = 1/8$, as with `balnet`. Both estimators share the weights, so a larger λ lowers the variance of both, and in every interior cell the variance falls at the path minimum. In an augmented estimator, however, the bias from the remaining imbalance scales with the error of the outcome model rather than with the outcome coefficients themselves (Athey et al. 2018). At full signal the optimal λ of ABW adds only 13% to 16% of the bias it adds to `balnet` for the linear and exponential outcomes.[10] Here, at $c \leq 0.5$ the variance saved outweighs the bias added, while at

[10] For the quadratic outcome at full signal and $c \leq 0.25$, a larger λ adds no bias to ABW, and at $c = 0.15$ lowers

$c = 1$ there is limited variance to remove. λ selection for ABW therefore pays wherever overlap is degraded.

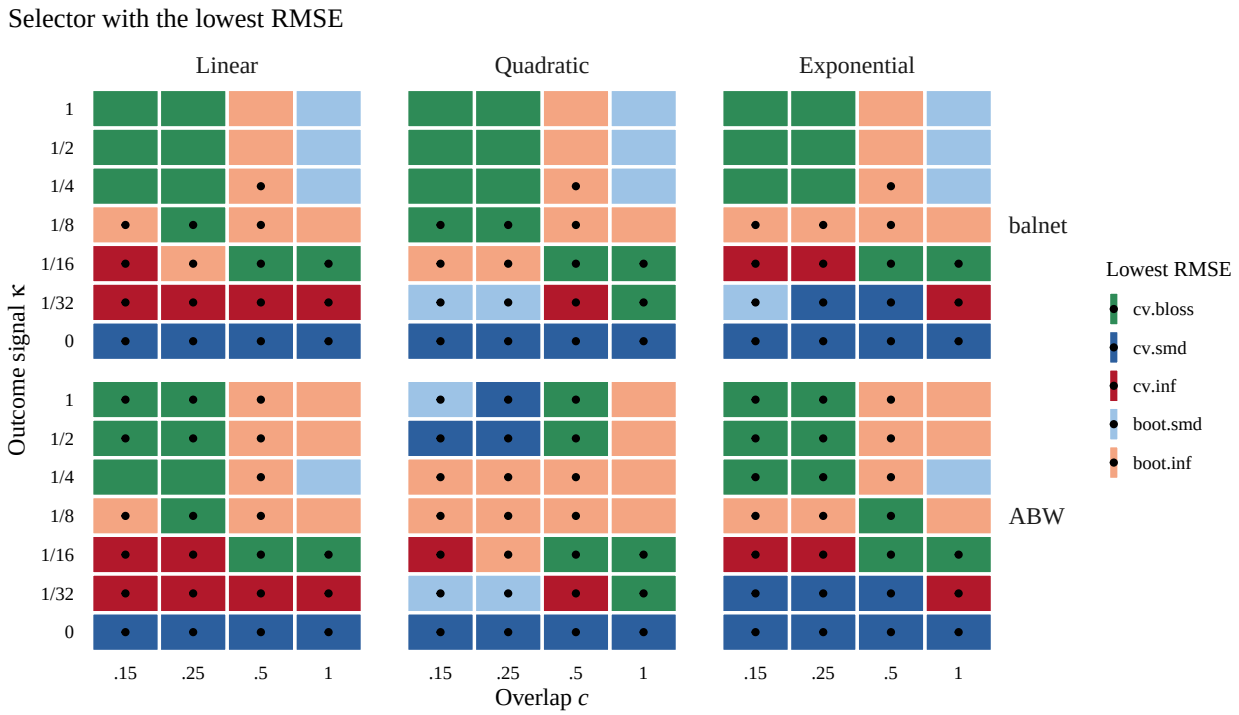


Figure 8: Selector with the lowest RMSE per cell by c and κ, $n = 1000$, $p = 50$. A dot marks a win against the endpoint.

Selector wins: Figure 8 marks the selector with the lowest RMSE in each cell and whether it beats the endpoint. At $\kappa = 0$ every selector improves on the endpoint of `balnet` at $c \leq 0.5$, and `cv.smd` has the lowest RMSE in every cell. As the signal grows the lead passes to `cv.bloss` and `boot.inf`, the only two still beating the untuned estimators in most cells at $\kappa = 1/8$. By $\kappa = 1/4$ only `boot.inf` wins, in the three cells at $c = 0.5$. This is consistent with findings from Tan (2017), where cross-validating this loss beat unpenalised calibration for pure noise but lost under the linear outcome we share, as the penalty it keeps biases the fit along the covariate the outcome loads on most. For ridge-augmented linear balancing weights, Bruns-Smith et al. (2026) found cross-validating imbalance the best selector and cross-validating the balancing loss itself the worst. Here, we find under lasso-penalised logistic weights the order reverses, with `cv.bloss` winning in all nine cells for ABW at full signal and $c \leq 0.5$ while `cv.smd` and `cv.inf` lead only with little or no signal. Whether the weights' form or the penalty drives the reversal is beyond our design.

The selectors use only the covariates and treatment, so the λ each picks does not change with the outcome signal. The RMSE-minimising λ, however, falls as the signal grows, approaching the endpoint for `balnet` and settling above it for ABW (Figure 7). Selector wins are therefore confined to the signal level where the RMSE-minimising λ passes the one a selector picks.

Scored by the bootstrap rather than cross-validation, the same imbalance criterion selects better balanced covariates. This is most pronounced under good overlap, where `boot.smd` and `boot.inf` set $\lambda \approx 0.01$ at $c = 1$ against 0.14 to 0.62 for `cv.smd` and `cv.inf` at every overlap. Under good

it, since balance on X does not guarantee unbiasedness for an outcome that is not a linear function of X (Zhao 2019).

overlap the weights near the endpoint are likely robust to resampling, while under poor overlap they rest on a few units with extreme weights, so the bootstrap selects a larger λ. Over all grids bootstrapping wins on `inf` in 52% of cells for `balnet` and 70% for ABW, against 43% and 55% for cross-validation, while on `smd` both methods perform similarly (Appendix Table 24). The difference lies mostly where the endpoint is optimal. There cross-validation performs far worse, with RMSE around three times the endpoint's for `balnet` at $c = 0.5$, while the bootstrap stays within about a tenth of it under good overlap. This suggests bootstrapping is the safer scoring across criteria, though `cv.bloss` remains the selector that wins most often.

In other settings: The remaining grids repeat this pattern (Appendix I). The penalty form matters mainly under poor overlap, where every $\alpha < 1$ improves the endpoint of ABW at every signal level while the lasso stays best for `balnet` at full signal (Appendix Table 25). Along n and p interior optima follow how far the path stops short of exact balance, rising with n at poor overlap and retreating to weak signal as p grows (Appendix Figure 7). Misspecification leaves `balnet` almost unchanged and removes most of ABW's full-signal optima, and in the denser grid ABW keeps a full-signal optimum in every cell only under equicorrelation (Appendix Figure 8). In no grid does a selector beat the endpoint of `balnet` at full signal under the lasso, and under good overlap neither estimator gains from approximate balance at full signal.

6 Conclusion

Table 11: $\bar{R}(\Theta)$ of A against B over every cell of each Study 1 grid, both specifications.

		$\bar{R}(\Theta)$, A vs B			
Grid	Cells	balnet vs glmnet (Hájek)	balnet vs AIPW-glmnet	ABW-balnet vs balnet	ABW-balnet vs AIPW-glmnet
Sample size	30	68.6	33.7	0.1	33.8
Overlap	84	59.9	23.1	20.2	32.5
Dimension	72	44.4	-26.4	33.3	15.7
Denser confounders	408	47.3	-3.5	17.4	14.5
Outcome signal	48	34.2	10.2	0.0	10.3

Largest MCSE: 0.5 points ($\bar{R}(\Theta)$). Cells left out: where A or B lacks more than 1% of its estimates; the pair's averages skip them. Cells: design cells × three outcomes × two specifications; on the denser grid the misspecification is Tan's transformation.

Across the comparable cells of Study 1, balancing weights have lower RMSE than MLE weights in nearly every setting, alone and augmented, under both propensity scores and every outcome function, with two exceptions. Firstly, in the eight noisiest cells under a correctly specified score, the Hájek estimator has lower RMSE than $\texttt{balnet}_{10^{-4}}$, as exact balance there costs more in variance than it removes in bias. Secondly, between the augmented estimators, AIPW is also level with or ahead of ABW where common support is almost certainly violated under a misspecified score and a nonlinear outcome. Elsewhere, $\texttt{balnet}_{10^{-4}}$ leads the Hájek estimator by between a third and two thirds on average across the grids (Table 11).

Augmentation narrows the lead, to under a fifth for ABW over AIPW on the dimension and denser grids, and misspecification widens ABW's lead on every grid but the denser one. Between the unaugmented $\texttt{balnet}_{10^{-4}}$ and AIPW, the performance ordering depends on whether the path reaches exact balance. Where it stops short, from $p/n = 0.5$ and at $c \leq 0.25$, AIPW is generally ahead. Where it reaches exact balance, $\texttt{balnet}_{10^{-4}}$ generally has lower RMSE, more so under

misspecification. Augmentation therefore restores the lead of balancing weights wherever exact balance fails.

As the sample grows, balancing weights stay ahead of MLE weights from $n = 250$ to 10^6, and the misspecified ordering Tan (2020b) report at $n \leq 800$ holds to $n = 10^6$. What changes with n is how each approach fails. The balancing weights converge to a fixed bias, which a misspecified outcome model cannot remove, as ABW reduces to OLS on $\tilde{X}$ at exact balance (Bruns-Smith et al. 2026). The MLE weights never settle, as their tails lengthen with n, extending to $n = 10^6$ the worsening Testa et al. (2026) saw at $n = 1000$. At the empirical SE this generally leaves the balancing estimators with no coverage by $n = 10{,}000$ and the MLE estimators with nominal coverage carried by their tails, so each fails to produce reliable intervals for different reasons.

Under degrading overlap ABW separates from $\texttt{balnet}_{10^{-4}}$ only where its path stops short, and more data pushes that point to poorer overlap. This is consistent with the reasoning Athey et al. (2018) provide for the lead of AIPW over their approximate balancing weights. We find under misspecification the lead of $\texttt{balnet}_{10^{-4}}$ over AIPW also extends to poorer overlap levels. Where common support is almost certainly violated, ABW keeps most of its coverage under a correctly specified score, while $\texttt{balnet}_{10^{-4}}$ and the Hájek estimator have none and AIPW keeps it only for the quadratic outcome. Regardless, nominal coverage does not make an estimator reliable in this setting. At the weakest overlap under misspecification no weighting estimator removes half the confounding, as in Keele et al. (2026). There the lowest RMSE does not improve with n. Here applied researchers may instead consider changing their estimand.

As more covariates are considered, `balnet` becomes an approximate balancing weight from $p/n = 0.5$. Here the error of $\texttt{balnet}_{10^{-4}}$ becomes bias-driven and its lead over the Hájek estimator narrows with p, consistent with the findings of Athey et al. (2018). Misspecification costs the MLE estimators less as p grows, so under it the lead of ABW over AIPW narrows, as the bias of ABW grows faster once the path stops short while the variance of AIPW falls. In Athey et al. (2018) that lead held steady under misspecification, but their models are misspecified through one covariate only, whereas we misspecify through every confounder in $\tilde{X}$. To our knowledge, no previous simulation has taken lasso-penalised balancing weights and cross-validated lasso MLE weights, alone and augmented, past $p = n$ under both a correct and a misspecified propensity score.

With the number of active confounders s increasing, unaugmented balancing weights keep their lead at every s, ν and covariance structure under both scores. Under independent and AR(1)-correlated covariates their lead widens with s while exact balance holds, as $\texttt{balnet}_{10^{-4}}$ stays close to unbiased and the error of the MLE weights grows with the number of active covariates (Tan 2020a). Once the path stops short, we find ABW's lead over AIPW grows with s and the unaugmented $\texttt{balnet}_{10^{-4}}$ catches AIPW by $s = 64$. Under equicorrelation, where every covariate shares a common factor, the MLE bias falls with s and the lead narrows at exact balance. Under Tan's transformation misspecification narrows the lead of balancing weights only while exact balance holds, as balancing a growing number of transformed covariates does not guarantee balance on the true ones.

As the outcome signal weakens, the ranking of the unaugmented estimators reverses, and in the noisiest cells under a correctly specified score $\texttt{balnet}_{10^{-4}}$ has higher RMSE than the Hájek estimator. Exact balance leaves it a smaller effective sample size, and the bias it removes falls with the signal,

so its efficiency advantage holds only once the signal clears the noise. Under misspecification it stays ahead at every noise level but the highest.

Our second study finds that approximate balance, a λ before the endpoint of `balnet`'s path, lowers RMSE below the endpoint in 61% of cells for `balnet` and 79% for ABW. We also find that tuning captures this gain in almost every such cell, though which selector does best differs across settings and between cross-validation and bootstrapping. For the unaugmented `balnet`, both findings hold under the Wong–Chan design, the interior optimum in fourteen of fifteen matched cells and the selector win in all fifteen (Appendix H). Under our design we observe for the unaugmented `balnet` that the outcome signal decides whether approximate balance is preferable, while overlap affects the size of its advantage. At full signal it is optimal in only 8% of cells, with its advantage reduced further where the path already stops short of exact balance. For ABW, overlap largely decides whether approximate balance is optimal, as the outcome model absorbs most of the bias a larger λ adds. Across penalty forms the lasso remains the better default for the unaugmented `balnet` at full signal, while ABW improves from the elastic net to ridge, $\alpha < 1$, as overlap degrades.

Of the five selectors, `cv.bloss` wins most often for both estimators, yet no selector maintains the lowest RMSE across more than around 30% of cells. When the endpoint is already optimal, tuning imbalance by bootstrap is a generally safer approach than by cross-validation. For naive application, where the outcome signal and the propensity score's specification are unknown, we recommend `boot.inf` for the unaugmented `balnet`, as it lowers RMSE below the path endpoint's by 5% on average across our settings and wins against it in 52% of cells. Where it loses, in 32% of cells, its RMSE sits on average a third above the endpoint's, at worst more than three times as high, in the denser setting under Tan's transformation with 64 confounders and at full signal. For ABW, `cv.bloss` and `boot.inf` are level on average, so `cv.bloss` remains our choice based on its win count.

Both studies hinge on how strong the outcome signal is against the noise. Where it is weak, every estimator's bias is small against the noise, so approximate balance lowers RMSE and coverage at the empirical SE stays near nominal under either score. As the signal grows, exact balance becomes the better choice for the unaugmented `balnet` and under misspecification `balnet`'s bias costs it coverage. In the worst case under a linear outcome, it holds nominal coverage only to $\kappa = 0.1$ and falls from there to 0.45 at $\kappa = 1$, where the bias is about twice the EmpSE and a correctly sized 95% confidence interval would exclude τ with around 55% probability. AIPW fares no better there, at 0.69 and 0.65 under the linear and exponential outcomes. For `balnet`, low RMSE with nominal coverage at full signal is only achievable under a correctly specified score. There both the unaugmented `balnet` and ABW keep nominal coverage at every κ and AIPW stays within a few points. Under our design, only the Hájek estimator loses its coverage, from $\kappa = 0.2$, as the lasso penalty shrinks its propensity score and that bias does not fall with n.

For applied researchers, who can generally observe overlap but not the outcome signal, our recommendations consider the naive application of `balnet` under situations where overlap may be weak but common support is likely to hold, and where we are certain overlap is strong. We ignore settings where there are likely violations, as in these settings a reliable estimate of the treatment effect may be beyond reach.

When confident that overlap is strong, the endpoint of ABW is the choice, as the path reaches exact balance, where it equals the unaugmented `balnet`, and pulls ahead of it wherever exact balance later fails under increased dimensionality or limited overlap.

Where overlap is reduced but common support is likely to hold, the path still reaches exact balance, so ABW at its endpoint still equals the unaugmented `balnet`, but tuning can provide benefits. Our study recommends that applied researchers using `balnet` implement ABW tuned by `cv.bloss`. However, our findings are unable to establish a universally best tuning approach, which remains an open question for future research.

6.1 Limitations

Our conclusions rest on one family of data-generating models. Within it, misspecification takes a different form on each grid, the Kang–Schafer transformations on the four-confounder grids and Tan's transformation on the denser grids. On the cells the grids share, the Kang–Schafer form costs the balancing and augmented estimators more for the linear outcome, so the two grids' misspecified results are not interchangeable. In Study 1, Tan's transformation also runs only at $c = 1$, so how it interacts with limited overlap is untested there.

Positivity holds by construction, so the violations of common support we report are inferred from how much of the score's distribution lies near the boundary rather than imposed. Our propensity coefficients are all positive and the misspecified balancing bias is negative in every cell, so its sign may differ under other coefficient patterns. The denser grids add confounders to the propensity score and the outcome together,[^12] so we cannot attribute the loss of AIPW as confounders are added to its propensity model alone.

Exact balance is generally feasible only when both arms have more units than p (Wager 2024), so with a treated share of one half it fails from $p/n = 0.5$, and rarer treatment would bring that point earlier. The denser grids use one correlation level, 0.5, and adjacent active covariates. They run 1000 replications rather than 10 000, and $s = 64$ is not run at $n = 500$ and $p/n = 0.1$.

The MLE weights are tuned by five-fold cross-validation of the prediction error, not of balance, and prediction-tuned lasso can over-regularise the propensity model (Wyss et al. 2026). Part of the advantage of balancing weights in high dimensions may then reflect this tuning rule rather than MLE itself. The MLE weights are also untrimmed, unlike the scores of Athey et al. (2018). $\texttt{balnet}_{10^{-4}}$ is taken at the end of its path rather than at a tuned λ, so Study 1 compares two default practices. Where the path stops short, the stopping rule is not logged, so infeasibility cannot be separated from slow convergence.

ABW and AIPW share one lasso outcome model, tuned by cross-validation and fitted on the same data as the weights without cross-fitting, so the gains from augmentation rest on that model. ABW with an unweighted outcome model also sits outside the model-assisted theory of Tan (2020a). No variance estimator is implemented for the balancing weights, so coverage is reported at the empirical SE only, the coverage of a correctly sized interval.

Study 2 runs 2500 replications per cell, 1000 at the two largest samples, so its interior optimum rule is a reporting threshold of about two Monte Carlo standard errors. The RMSE-minimising λ is chosen on the replications it is scored on, so its advantage over the endpoint is optimistic.

The selector averages weight each setting by how often our grids run it, so another mix of signal, overlap and penalty form would move them and could change our recommendation. Our selectors never see the outcome, so the outcome-based tuning that Bruns-Smith et al. (2026) recommend as a starting point for ridge-augmented weights is untested. The selectors use ten folds throughout, with no other fold count tested. Under ridge and the elastic net the endpoint differs from the lasso's by construction, so the endpoints are not like for like. The Wong–Chan check uses `balnet` alone, the ATT and 1000 replications.

Future work could add a variance estimator and the coverage of its intervals, tune the MLE weights by balance rather than prediction, record why each path stops, and vary the treated share.

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Appendix

Appendix A: Correct specification

Appendix Table 1: Per cent lower RMSE $\bar{R}(\Theta)$ of A against B under correct specification, $n = 500$ to 400k.

		$\bar{R}(\Theta)$			
Estimator (A)	Comparator (B)	All	Linear	Quadratic	Exponential
balnet	glmnet (Hájek)	71.7	77.0	65.3	71.4
ABW-balnet	AIPW-glmnet	4.8	-0.2	11.0	3.2
AIPW-glmnet	glmnet (Hájek)	70.3	77.1	61.1	70.6

Largest MCSE: 0.1 points ($\bar{R}(\Theta)$). The averages are over 24 cells, 8 per outcome, at 10 000 Monte Carlo replications.

Appendix B: Misspecification

Fit diagnostics

Appendix Table 2: Fit diagnostics of the glmnet and Oracle weights under a misspecified propensity score, linear outcome. Median (lower–upper quartile) across replications.

		n (thousands)	
		0.5	1
ESS			
glmnet	Treated	215 (198–228)	396 (346–425)
	Control	231 (222–240)	454 (441–467)
Oracle	Treated	193 (182–203)	383 (367–399)
	Control	194 (182–203)	383 (367–398)
Largest weight			
glmnet	Treated	8.5 (6.4–12.2)	14.3 (10.2–22.4)
	Control	4.3 (3.8–5.1)	5.1 (4.5–5.9)
Oracle	Treated	9.3 (7.8–11.6)	11.2 (9.3–13.9)
	Control	9.3 (7.7–11.6)	11.2 (9.5–13.7)
Max |SMD|			
glmnet	Treated	0.123 (0.107–0.145)	0.095 (0.082–0.118)
	Control	0.104 (0.094–0.114)	0.080 (0.073–0.088)
Oracle	Treated	0.144 (0.127–0.166)	0.103 (0.091–0.118)
	Control	0.142 (0.126–0.163)	0.101 (0.090–0.115)

No MCSE, as these describe the fits. ESS: Kish's, per arm. Largest weight: the arm's weights scaled to sum to n. Max |SMD|: over the covariates the fits use ($\tilde{X}$), weighted arm mean against the full-sample mean in full-sample SDs. Oracle: weights from the true propensity score. The weights do not depend on the outcome.

Sample sizes up to 1M

Appendix Table 3: RMSE (bias) under a misspecified propensity score, by n. n = 1M is a separate run.

n	balnet	ABW-balnet	glmnet (Hájek)	glmnet (HT)	AIPW-glmnet	ABW-balnet (true OR)	AIPW-glmnet (true OR)
Linear							
250	0.332 (−0.264)	0.289 (−0.205)	0.634 (−0.610)	0.598 (−0.574)	0.347 (−0.301)	0.190 (−0.064)	0.201 (−0.133)
500	0.227 (−0.181)	0.226 (−0.181)	0.495 (−0.472)	0.479 (−0.450)	0.302 (−0.268)	0.123 (−0.030)	0.130 (−0.073)
1k	0.205 (−0.186)	0.205 (−0.185)	0.399 (−0.364)	0.934 (−0.338)	0.314 (−0.270)	0.078 (−0.021)	0.091 (−0.040)
2k	0.201 (−0.191)	0.201 (−0.191)	0.332 (−0.275)	2.452 (−0.228)	0.388 (−0.295)	0.055 (−0.015)	0.071 (−0.023)
3k	0.201 (−0.195)	0.201 (−0.194)	0.303 (−0.235)	0.696 (−0.209)	0.428 (−0.314)	0.045 (−0.012)	0.065 (−0.017)
4k	0.202 (−0.197)	0.201 (−0.196)	0.305 (−0.203)	7.818 (−0.049)	0.503 (−0.335)	0.039 (−0.011)	0.066 (−0.013)
5k	0.202 (−0.198)	0.202 (−0.198)	0.281 (−0.188)	1.468 (−0.144)	0.499 (−0.341)	0.035 (−0.010)	0.063 (−0.013)
10k	0.204 (−0.202)	0.204 (−0.202)	0.310 (−0.121)	12.304 (0.086)	0.638 (−0.395)	0.024 (−0.007)	0.071 (−0.005)
15k	0.206 (−0.205)	0.206 (−0.204)	0.284 (−0.100)	22.247 (0.213)	0.701 (−0.418)	0.020 (−0.006)	0.065 (−0.006)
20k	0.207 (−0.206)	0.207 (−0.206)	0.293 (−0.080)	477 (4.924)	0.768 (−0.441)	0.017 (−0.005)	0.076 (−0.003)
50k	0.210 (−0.210)	0.210 (−0.209)	0.343 (−0.027)	19.774 (0.437)	0.899 (−0.505)	0.011 (−0.003)	0.075 (0.000)
75k	0.212 (−0.212)	0.212 (−0.211)	0.338 (−0.001)	3.521 (0.162)	1.003 (−0.558)	0.009 (−0.003)	0.080 (−0.002)
100k	0.213 (−0.212)	0.212 (−0.212)	0.364 (0.011)	127 (1.836)	1.079 (−0.579)	0.008 (−0.002)	0.082 (−0.002)
200k	0.215 (−0.214)	0.214 (−0.214)	0.375 (0.045)	11.389 (0.346)	1.149 (−0.641)	0.006 (−0.002)	0.085 (0.001)
1M	0.219 (−0.218)	0.218 (−0.218)	0.471 (0.100)	128 (2.307)	1.640 (−0.825)	0.003 (−0.001)	0.098 (−0.001)
Quadratic							
250	0.408 (−0.248)	0.376 (−0.190)	0.686 (−0.612)	0.663 (−0.572)	0.414 (−0.290)	0.313 (−0.016)	0.335 (−0.141)
500	0.257 (−0.137)	0.257 (−0.136)	0.549 (−0.413)	0.642 (−0.363)	0.310 (−0.166)	0.220 (0.063)	0.330 (0.038)
1k	0.183 (−0.119)	0.183 (−0.119)	0.556 (−0.246)	4.161 (−0.127)	0.300 (−0.086)	0.166 (0.094)	0.473 (0.168)
2k	0.144 (−0.105)	0.143 (−0.104)	0.677 (−0.088)	12.182 (0.190)	0.357 (−0.018)	0.153 (0.118)	0.667 (0.285)
3k	0.127 (−0.099)	0.127 (−0.098)	0.717 (−0.013)	3.531 (0.183)	0.385 (0.011)	0.152 (0.129)	0.725 (0.339)
4k	0.117 (−0.094)	0.117 (−0.094)	0.873 (0.058)	41.335 (0.959)	0.462 (0.043)	0.154 (0.137)	0.874 (0.393)
5k	0.112 (−0.093)	0.112 (−0.093)	0.815 (0.081)	8.604 (0.407)	0.449 (0.050)	0.153 (0.140)	0.832 (0.407)
10k	0.097 (−0.086)	0.097 (−0.086)	1.137 (0.244)	53.482 (1.307)	0.624 (0.125)	0.161 (0.154)	1.131 (0.535)
15k	0.092 (−0.084)	0.091 (−0.084)	1.144 (0.294)	151 (2.380)	0.617 (0.143)	0.163 (0.158)	1.155 (0.572)
20k	0.088 (−0.082)	0.087 (−0.081)	1.248 (0.349)	2274 (24.486)	0.662 (0.168)	0.166 (0.163)	1.265 (0.617)
50k	0.081 (−0.078)	0.081 (−0.078)	1.520 (0.500)	87.516 (2.835)	0.826 (0.238)	0.173 (0.171)	1.491 (0.738)
75k	0.078 (−0.077)	0.078 (−0.076)	1.611 (0.597)	22.318 (1.668)	0.851 (0.283)	0.176 (0.174)	1.597 (0.821)
100k	0.078 (−0.076)	0.077 (−0.076)	1.741 (0.635)	610 (9.788)	0.913 (0.299)	0.177 (0.176)	1.699 (0.853)
200k	0.076 (−0.075)	0.075 (−0.074)	1.833 (0.757)	57.201 (2.581)	0.955 (0.360)	0.180 (0.179)	1.807 (0.960)
1M	0.073 (−0.072)	0.072 (−0.072)	2.501 (1.053)	738 (13.956)	1.263 (0.510)	0.185 (0.185)	2.418 (1.234)
Exponential							
250	0.298 (−0.215)	0.271 (−0.175)	0.513 (−0.481)	0.507 (−0.447)	0.314 (−0.260)	0.194 (−0.039)	0.204 (−0.118)
500	0.199 (−0.143)	0.199 (−0.143)	0.404 (−0.365)	0.457 (−0.294)	0.238 (−0.200)	0.127 (0.002)	0.145 (−0.034)
1k	0.165 (−0.140)	0.165 (−0.140)	0.348 (−0.271)	2.550 (−0.121)	0.200 (−0.173)	0.082 (0.015)	0.167 (0.022)
2k	0.151 (−0.139)	0.151 (−0.139)	0.337 (−0.189)	7.027 (0.093)	0.183 (−0.159)	0.062 (0.023)	0.220 (0.065)
3k	0.147 (−0.138)	0.147 (−0.138)	0.330 (−0.151)	2.005 (0.103)	0.177 (−0.155)	0.054 (0.028)	0.233 (0.085)
4k	0.144 (−0.138)	0.144 (−0.138)	0.385 (−0.117)	24.120 (0.571)	0.179 (−0.152)	0.051 (0.031)	0.291 (0.103)
5k	0.143 (−0.138)	0.143 (−0.138)	0.347 (−0.104)	4.827 (0.250)	0.178 (−0.152)	0.048 (0.032)	0.265 (0.107)
10k	0.140 (−0.137)	0.140 (−0.137)	0.475 (−0.028)	33.084 (0.823)	0.194 (−0.143)	0.046 (0.038)	0.387 (0.151)
15k	0.140 (−0.138)	0.139 (−0.138)	0.462 (−0.005)	86.034 (1.415)	0.185 (−0.146)	0.045 (0.040)	0.388 (0.162)
20k	0.139 (−0.137)	0.139 (−0.137)	0.511 (0.020)	1407 (15.142)	0.183 (−0.144)	0.046 (0.041)	0.446 (0.178)
50k	0.138 (−0.137)	0.138 (−0.137)	0.632 (0.088)	53.357 (1.764)	0.209 (−0.139)	0.047 (0.045)	0.527 (0.219)
75k	0.138 (−0.137)	0.138 (−0.137)	0.650 (0.128)	12.766 (1.047)	0.192 (−0.141)	0.048 (0.047)	0.556 (0.246)
100k	0.138 (−0.137)	0.137 (−0.137)	0.719 (0.146)	352 (5.784)	0.209 (−0.142)	0.048 (0.047)	0.602 (0.257)
200k	0.138 (−0.137)	0.137 (−0.137)	0.755 (0.200)	34.525 (1.603)	0.201 (−0.138)	0.050 (0.049)	0.650 (0.296)
1M	0.137 (−0.137)	0.137 (−0.137)	1.071 (0.329)	447 (8.427)	0.233 (−0.129)	0.052 (0.052)	0.910 (0.400)

Largest MCSE: 1100 (RMSE; glmnet (HT), quadratic, n = 20k) and 23 (bias; glmnet (HT), quadratic, n = 20k). Values of 100 or more are shown to the unit. Without glmnet (HT): largest MCSE 0.089 (RMSE) and 0.023 (bias).

Averages

Appendix Table 4: Per cent lower RMSE $\bar{R}(\Theta)$ of A against B under a misspecified propensity score, n = 500 to 50k. Other pairs on these cells are in Table 4 and Appendix Table 5.

		$\bar{R}(\Theta)$			
Estimator (A)	Comparator (B)	All	Linear	Quadratic	Exponential
ABW-balnet	glmnet (HT)	96.9	91.3	98.6	97.4
ABW-balnet (true OR)	AIPW-glmnet (true OR)	69.9	52.0	76.6	75.7

Largest MCSE: 1.1 points ($\bar{R}(\Theta)$). True OR: the outcome model fitted on the true covariates X. The averages are over 15 cells, 5 per outcome, at 10 000 Monte Carlo replications.

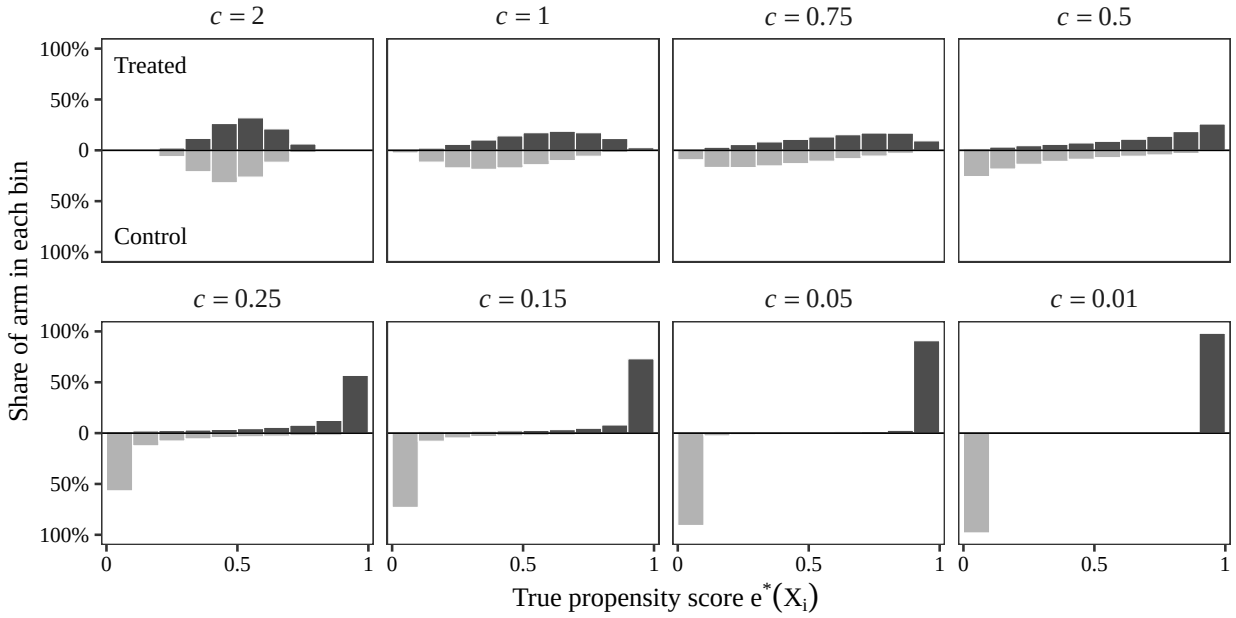
Appendix Table 5: Per cent lower RMSE $\bar{R}(\Theta)$ and share of bias $\bar{s}_m(\Theta)$ of balnet against Oracle (Hájek) and OLS, n = 500 to 50k.

		$\bar{R}(\Theta)$				$\bar{s}_m(\Theta)$	
Estimator (A)	Comparator (B)	All	Linear	Quadratic	Exponential	A	B
Both specifications							
balnet	Oracle (Hájek)	-13.8	-27.5	7.2	-24.7	0.39	0.00
	OLS	6.1	1.1	12.4	4.5	0.39	0.46
Correct specification							
balnet	Oracle (Hájek)	38.6	46.4	32.3	36.4	0.00	0.00
	OLS	-0.4	-6.7	9.5	-4.6	0.00	0.10
Misspecified propensity score							
balnet	Oracle (Hájek)	-111.2	-203.2	-27.1	-144.3	0.78	0.00
	OLS	12.2	8.4	15.3	12.8	0.78	0.82

Largest MCSE: 1.0 points ($\bar{R}(\Theta)$), 0.002 ($\bar{s}_m(\Theta)$). The averages are over 30 cells, 15 per specification, at 10 000 Monte Carlo replications.

Appendix C: Overlap

Overlap levels



Appendix Figure 1: True propensity score by arm at each overlap level c, as the share of each arm in bins of width 0.1. The outermost bins hold the units outside [0.1, 0.9].

Appendix Table 6: Share of units with a true propensity score $e^*(X_i)$ outside each interval, by overlap level c.

	Outside		
c	[0.1, 0.9]	[0.05, 0.95]	[0.01, 0.99]
2	< 0.1%	< 0.1%	< 0.1%
1	2.8%	0.3%	< 0.1%
0.75	9.9%	2.7%	0.1%
0.5	27.2%	14.1%	2.2%
0.25	58.3%	46.2%	25.1%
0.15	74.2%	65.9%	49.1%
0.05	91.3%	88.3%	81.8%
0.01	98.2%	97.7%	96.3%

c = 0.01 is used in the $n \times c$ grid only.

Performance

Appendix Table 7: Bias and RMSE under correct specification, linear outcome, n = 1000 and 10 000.

		n = 1000			n = 10000		
	Overlap c	0.15	0.25	0.5	0.15	0.25	0.5
balnet	Bias	−0.699	−0.377	−0.013	−0.303	−0.002	−0.001
	RMSE	0.725	0.414	0.126	0.339	0.095	0.036
glmnet (Hájek)	Bias	−1.326	−0.877	−0.452	−0.766	−0.380	−0.155
	RMSE	1.348	0.922	0.476	0.856	0.480	0.177

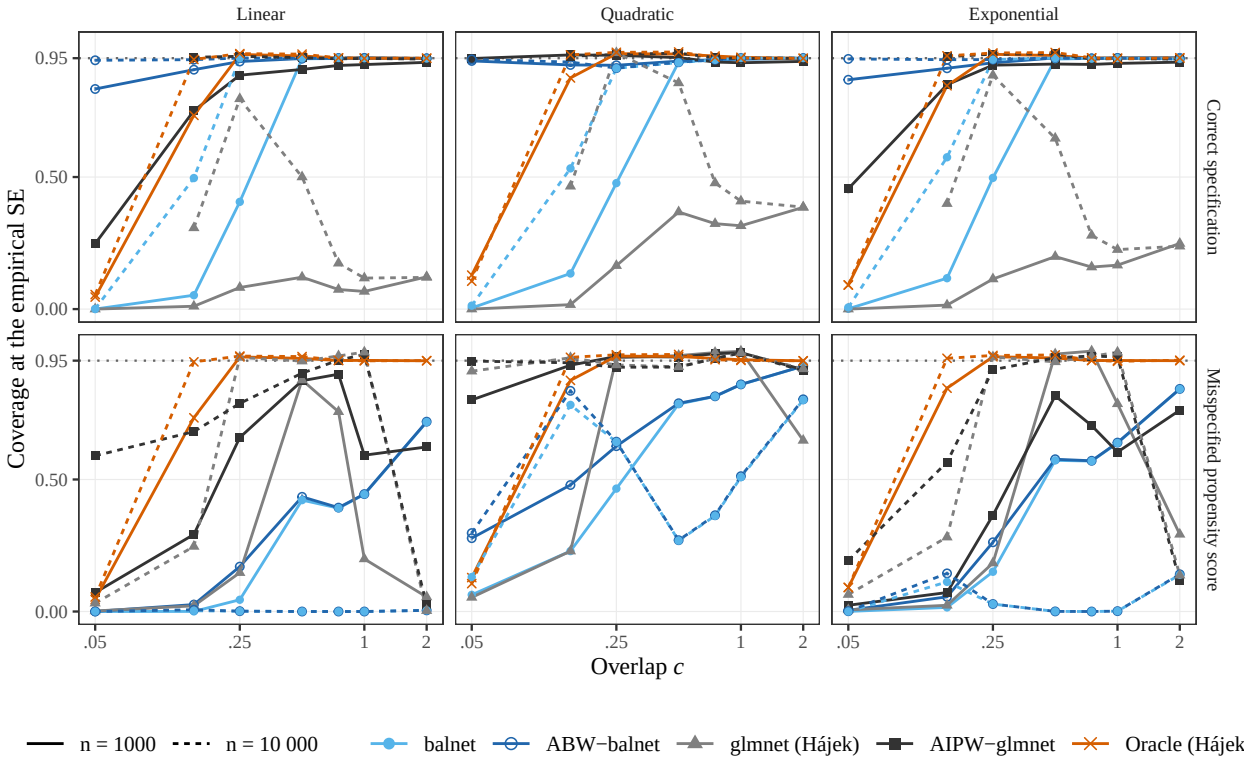
Largest MCSE: 0.0038 (Bias), 0.0033 (RMSE). No estimate is missing. τ = 0. balnet paths ending above λ = 1.01e-4: in 5 of 6 cells, up to 10 000 (treated) and 10 000 (control) per cell.

Appendix Table 8: RMSE under a misspecified propensity score, n = 1000 and 10 000.

		n = 1000		n = 10000	
	Overlap c	0.05	0.15	0.05	0.15
Linear					
balnet	RMSE	1.376	1.022	1.316	0.753
ABW-balnet	RMSE	1.089	0.809	1.154	0.706
glmnet (Hájek)	RMSE	1.676	1.382	1.565	1.108
AIPW-glmnet	RMSE	1.159	1.039	1.970	1.751
Quadratic					
balnet	RMSE	1.458	1.038	1.285	0.471
ABW-balnet	RMSE	1.163	0.831	1.084	0.425
glmnet (Hájek)	RMSE	1.724	1.453	1.413	1.207
AIPW-glmnet	RMSE	0.997	0.931	1.148	1.225
Exponential					
balnet	RMSE	1.193	0.873	1.102	0.558
ABW-balnet	RMSE	1.049	0.766	0.999	0.528
glmnet (Hájek)	RMSE	1.384	1.160	1.365	1.005
AIPW-glmnet	RMSE	1.002	0.863	1.093	0.910

Largest MCSE: 0.022 (RMSE). No estimate is missing. $\tau = 0$.

Coverage

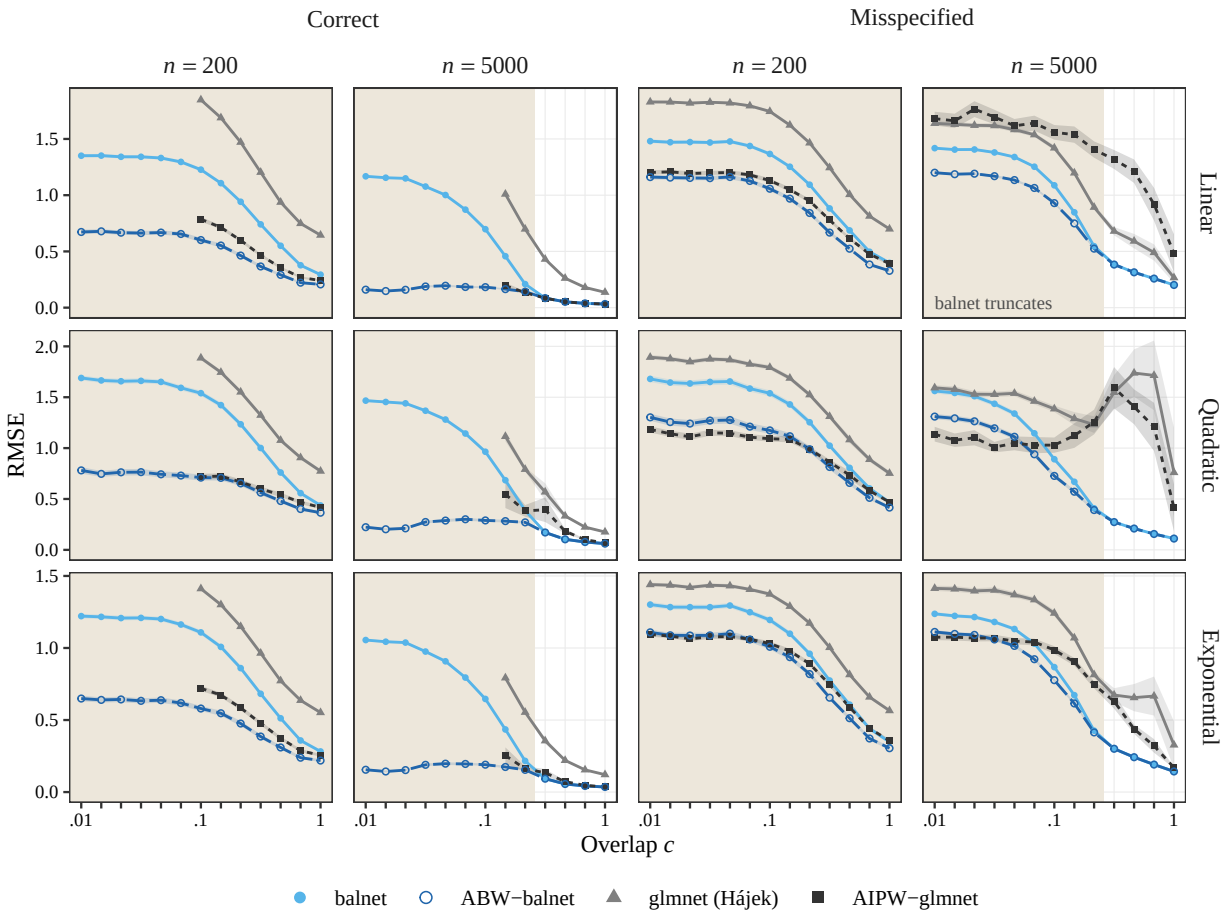


Appendix Figure 2: Coverage at the empirical SE by overlap c at n = 1000 and 10 000, both specifications. The dotted line is the nominal 95%.

Appendix D: Overlap and sample size

Performance

RMSE against overlap c



Appendix Figure 3: RMSE against overlap $c = 0.01$ to 1 at $n = 200$ and 5000, both specifications. Shading marks where the balnet path truncates in at least half the replications (Appendix Table 10). Modelling estimators are not drawn where more than 1% of replications have undefined weights.

Appendix Table 9: RMSE under a misspecified propensity score at the two weakest overlaps.

		Overlap c	
		0.01	0.0147
Linear · n = 200			
balnet	RMSE	1.480	1.471
ABW-balnet	RMSE	1.161	1.156
glmnet (Hájek)	RMSE	1.829	1.828
AIPW-glmnet	RMSE	1.202	1.208
Linear · n = 5000			
balnet	RMSE	1.418	1.405
ABW-balnet	RMSE	1.200	1.187
glmnet (Hájek)	RMSE	1.639	1.630
AIPW-glmnet	RMSE	1.681	1.661
Quadratic · n = 200			
balnet	RMSE	1.680	1.644

		Overlap c	
		0.01	0.0147
ABW-balnet	RMSE	1.302	1.256
glmnet (Hájek)	RMSE	1.893	1.878
AIPW-glmnet	RMSE	1.183	1.141
Quadratic · n = 5000			
balnet	RMSE	1.562	1.542
ABW-balnet	RMSE	1.310	1.293
glmnet (Hájek)	RMSE	1.593	1.577
AIPW-glmnet	RMSE	1.137	1.078
Exponential · n = 200			
balnet	RMSE	1.301	1.284
ABW-balnet	RMSE	1.109	1.088
glmnet (Hájek)	RMSE	1.440	1.436
AIPW-glmnet	RMSE	1.094	1.082
Exponential · n = 5000			
balnet	RMSE	1.237	1.223
ABW-balnet	RMSE	1.111	1.098
glmnet (Hájek)	RMSE	1.414	1.410
AIPW-glmnet	RMSE	1.075	1.075

Largest MCSE: 0.037 (RMSE). No estimate is missing. $\tau = 0$.

Path truncation

Appendix Table 10: Replications (%) in which the balnet path truncates in at least one arm.

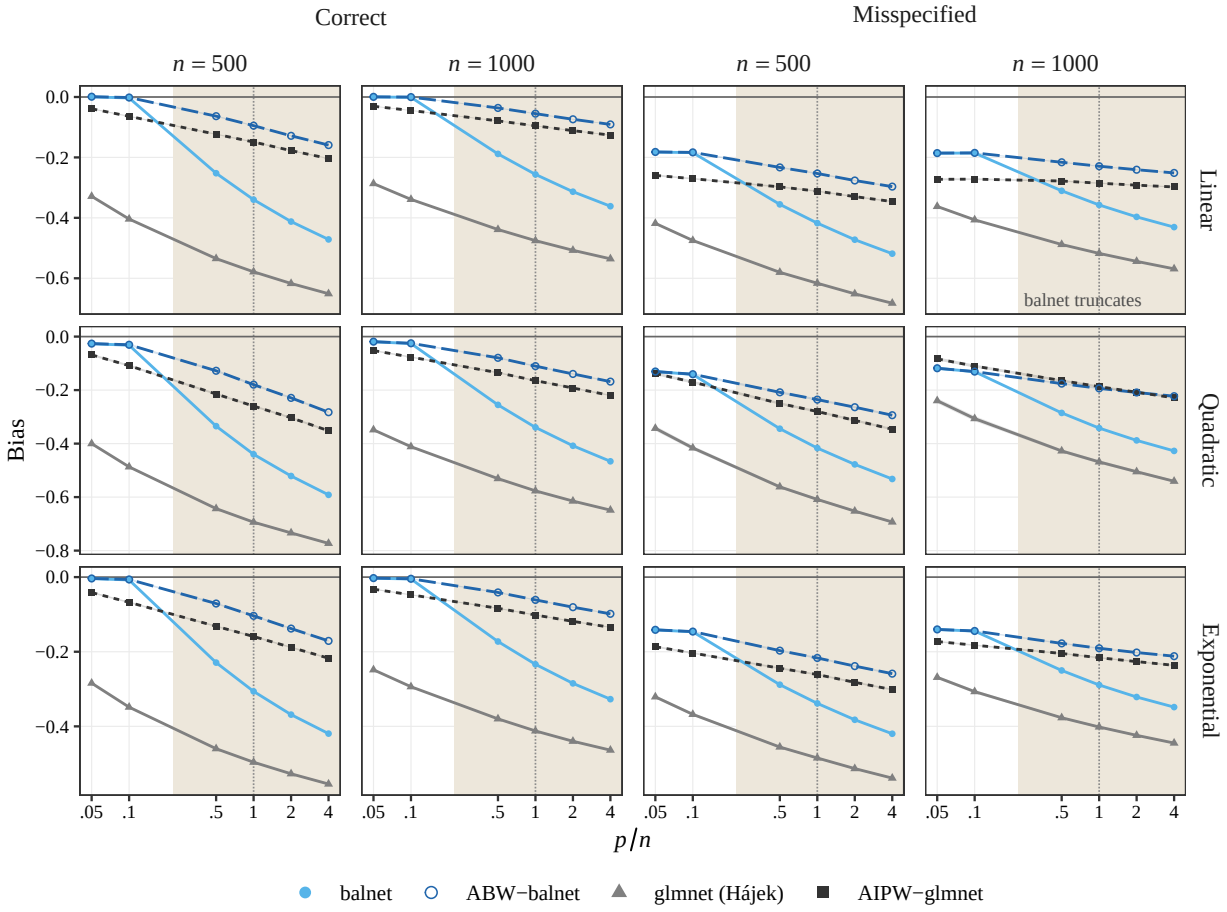
	Overlap c												
n	0.01	0.0147	0.0215	0.0316	0.0464	0.0681	0.1	0.147	0.215	0.316	0.464	0.681	1
Correct specification													
140	100	100	100	100	100	100	100	100	100	100	100	100	100
200	100	100	100	100	100	100	100	100	100	100	100	100	100
290	100	100	100	100	100	100	100	100	100	100	100	100	99
3500	100	100	100	100	100	100	100	100	100	31	0	0	0
5000	100	100	100	100	100	100	100	100	100	0	0	0	0
Misspecified propensity score													
140	100	100	100	100	100	100	100	100	100	100	100	100	100
200	100	100	100	100	100	100	100	100	100	100	100	100	100
290	100	100	100	100	100	100	100	100	100	100	100	100	98
3500	100	100	100	100	100	100	100	100	100	4	0	0	0
5000	100	100	100	100	100	100	100	100	96	0	0	0	0

Truncated: the path ends above λ = 1.01e-4. Largest MCSE: 1.5 percentage points.

Appendix E: High-dimensional settings

Performance

Bias against p/n



Appendix Figure 4: Bias against p/n at $n = 500$ and 1000, both specifications. Shading and dotted line as in Figure 4.

Appendix Table 11: Bias and RMSE under correct specification, by outcome.

		p/n				
		0.1	0.5	1	2	4
Linear · n = 500						
balnet	Bias	−0.003	−0.252	−0.340	−0.412	−0.471
	RMSE	0.121	0.276	0.357	0.426	0.483
ABW-balnet	Bias	−0.002	−0.063	−0.095	−0.129	−0.159
	RMSE	0.121	0.129	0.143	0.166	0.190
glmnet (Hájek)	Bias	−0.404	−0.535	−0.579	−0.617	−0.651
	RMSE	0.420	0.546	0.589	0.627	0.660
AIPW-glmnet	Bias	−0.064	−0.123	−0.149	−0.177	−0.204
	RMSE	0.120	0.160	0.180	0.205	0.228
Oracle (Hájek)	Bias	−0.012	−0.008	−0.006	−0.005	−0.008
	RMSE	0.198	0.196	0.199	0.199	0.200
Linear · n = 1000						
balnet	Bias	−0.001	−0.188	−0.256	−0.314	−0.361
	RMSE	0.083	0.204	0.267	0.322	0.369
ABW-balnet	Bias	0.000	−0.036	−0.055	−0.074	−0.091

		p/*n*				
		0.1	0.5	1	2	4
	RMSE	0.083	0.086	0.094	0.105	0.117
glmnet (Hájek)	Bias	−0.339	−0.439	−0.475	−0.507	−0.536
	RMSE	0.348	0.445	0.481	0.513	0.541
AIPW-glmnet	Bias	−0.045	−0.079	−0.096	−0.111	−0.126
	RMSE	0.084	0.106	0.119	0.132	0.145
Oracle (Hájek)	Bias	−0.004	−0.004	−0.005	−0.005	−0.003
	RMSE	0.142	0.139	0.141	0.142	0.142
Quadratic · n = 500						
balnet	Bias	−0.032	−0.335	−0.440	−0.521	−0.592
	RMSE	0.221	0.390	0.480	0.555	0.621
ABW-balnet	Bias	−0.030	−0.128	−0.180	−0.229	−0.283
	RMSE	0.221	0.239	0.265	0.300	0.340
glmnet (Hájek)	Bias	−0.487	−0.643	−0.694	−0.734	−0.773
	RMSE	0.527	0.671	0.719	0.758	0.796
AIPW-glmnet	Bias	−0.109	−0.215	−0.260	−0.304	−0.352
	RMSE	0.230	0.289	0.323	0.360	0.401
Oracle (Hájek)	Bias	−0.012	−0.011	−0.007	−0.001	−0.008
	RMSE	0.299	0.296	0.300	0.306	0.301
Quadratic · n = 1000						
balnet	Bias	−0.026	−0.255	−0.339	−0.408	−0.466
	RMSE	0.154	0.291	0.365	0.429	0.484
ABW-balnet	Bias	−0.025	−0.079	−0.110	−0.140	−0.168
	RMSE	0.153	0.163	0.175	0.195	0.215
glmnet (Hájek)	Bias	−0.412	−0.531	−0.577	−0.615	−0.649
	RMSE	0.435	0.548	0.592	0.629	0.662
AIPW-glmnet	Bias	−0.077	−0.135	−0.165	−0.192	−0.220
	RMSE	0.162	0.193	0.214	0.237	0.259
Oracle (Hájek)	Bias	−0.005	−0.003	−0.006	−0.006	−0.004
	RMSE	0.213	0.212	0.213	0.216	0.214
Exponential · n = 500						
balnet	Bias	−0.008	−0.229	−0.306	−0.369	−0.419
	RMSE	0.130	0.259	0.327	0.386	0.435
ABW-balnet	Bias	−0.006	−0.071	−0.104	−0.138	−0.171
	RMSE	0.130	0.140	0.155	0.179	0.205
glmnet (Hájek)	Bias	−0.349	−0.460	−0.496	−0.527	−0.555
	RMSE	0.368	0.474	0.509	0.539	0.567
AIPW-glmnet	Bias	−0.068	−0.132	−0.159	−0.188	−0.217
	RMSE	0.131	0.172	0.194	0.219	0.244
Oracle (Hájek)	Bias	−0.009	−0.006	−0.004	−0.003	−0.005
	RMSE	0.182	0.181	0.183	0.184	0.184
Exponential · n = 1000						
balnet	Bias	−0.005	−0.173	−0.234	−0.285	−0.327
	RMSE	0.090	0.192	0.247	0.295	0.336
ABW-balnet	Bias	−0.004	−0.041	−0.061	−0.081	−0.098
	RMSE	0.090	0.094	0.102	0.114	0.126
glmnet (Hájek)	Bias	−0.294	−0.380	−0.412	−0.440	−0.464
	RMSE	0.305	0.388	0.420	0.447	0.471
AIPW-glmnet	Bias	−0.048	−0.084	−0.102	−0.119	−0.135
	RMSE	0.092	0.114	0.128	0.142	0.156
Oracle (Hájek)	Bias	−0.003	−0.002	−0.004	−0.004	−0.002
	RMSE	0.131	0.129	0.130	0.131	0.131

Largest MCSE: 0.0031 (Bias), 0.004 (RMSE). No estimate is missing. $\tau = 0$.

Appendix Table 12: Bias and RMSE under a misspecified propensity score, by outcome.

		p/*n*				
		0.1	0.5	1	2	4
Linear · n = 500						
balnet	Bias	−0.184	−0.355	−0.417	−0.472	−0.519
	RMSE	0.229	0.377	0.435	0.487	0.531
ABW-balnet	Bias	−0.183	−0.233	−0.253	−0.276	−0.297
	RMSE	0.229	0.267	0.283	0.302	0.320
glmnet (Hájek)	Bias	−0.475	−0.580	−0.616	−0.651	−0.683
	RMSE	0.500	0.594	0.628	0.662	0.692
AIPW-glmnet	Bias	−0.270	−0.297	−0.312	−0.330	−0.346
	RMSE	0.306	0.323	0.335	0.350	0.366
Linear · n = 1000						
balnet	Bias	−0.186	−0.310	−0.357	−0.397	−0.431
	RMSE	0.209	0.323	0.368	0.405	0.438
ABW-balnet	Bias	−0.185	−0.216	−0.229	−0.241	−0.251
	RMSE	0.208	0.235	0.246	0.256	0.266
glmnet (Hájek)	Bias	−0.407	−0.488	−0.518	−0.544	−0.569
	RMSE	0.435	0.500	0.530	0.554	0.576
AIPW-glmnet	Bias	−0.272	−0.278	−0.285	−0.292	−0.298
	RMSE	0.312	0.299	0.305	0.309	0.312
Quadratic · n = 500						
balnet	Bias	−0.141	−0.344	−0.417	−0.478	−0.532
	RMSE	0.260	0.401	0.461	0.515	0.565
ABW-balnet	Bias	−0.141	−0.208	−0.236	−0.264	−0.294
	RMSE	0.259	0.296	0.313	0.332	0.354
glmnet (Hájek)	Bias	−0.416	−0.562	−0.609	−0.652	−0.693
	RMSE	0.587	0.627	0.657	0.692	0.725
AIPW-glmnet	Bias	−0.170	−0.250	−0.280	−0.313	−0.346
	RMSE	0.314	0.340	0.358	0.379	0.405
Quadratic · n = 1000						
balnet	Bias	−0.131	−0.285	−0.342	−0.388	−0.427
	RMSE	0.202	0.321	0.369	0.412	0.448
ABW-balnet	Bias	−0.131	−0.176	−0.194	−0.209	−0.223
	RMSE	0.201	0.231	0.241	0.253	0.263
glmnet (Hájek)	Bias	−0.307	−0.427	−0.468	−0.505	−0.541
	RMSE	0.595	0.523	0.566	0.580	0.583
AIPW-glmnet	Bias	−0.110	−0.164	−0.187	−0.208	−0.229
	RMSE	0.303	0.270	0.284	0.291	0.290
Exponential · n = 500						
balnet	Bias	−0.147	−0.288	−0.338	−0.382	−0.420
	RMSE	0.201	0.316	0.361	0.401	0.436
ABW-balnet	Bias	−0.146	−0.197	−0.217	−0.239	−0.259
	RMSE	0.201	0.238	0.252	0.269	0.286
glmnet (Hájek)	Bias	−0.368	−0.455	−0.485	−0.513	−0.539
	RMSE	0.415	0.477	0.502	0.529	0.552
AIPW-glmnet	Bias	−0.203	−0.244	−0.261	−0.282	−0.301
	RMSE	0.240	0.274	0.288	0.307	0.325
Exponential · n = 1000						
balnet	Bias	−0.145	−0.250	−0.289	−0.321	−0.348
	RMSE	0.174	0.266	0.302	0.332	0.358

		p/*n*				
		0.1	0.5	1	2	4
ABW-balnet	Bias	−0.144	−0.178	−0.191	−0.202	−0.212
	RMSE	0.174	0.201	0.211	0.220	0.229
glmnet (Hájek)	Bias	−0.307	−0.377	−0.402	−0.424	−0.445
	RMSE	0.377	0.401	0.428	0.444	0.458
AIPW-glmnet	Bias	−0.183	−0.205	−0.216	−0.226	−0.236
	RMSE	0.210	0.225	0.236	0.244	0.253

Largest MCSE: 0.0051 (Bias), 0.051 (RMSE). No estimate is missing. $\tau = 0$.

Fit diagnostics

Appendix Table 13: Nonzero coefficients of balnet and the glmnet propensity score under correct specification, linear outcome. Median (lower–upper quartile) across replications.

		p/*n*				
		0.1	0.5	1	2	4
Nonzero coefficients · n = 500						
glmnet PS	Both	10 (7–14)	12 (7–19)	12 (7–21)	12 (6–23)	12 (6–23)
balnet	Treated	50 (50–50)	102 (94–109)	109 (99–120)	111 (98–127)	109 (92–129)
	Control	50 (50–50)	101 (94–109)	109 (99–119)	110 (97–126)	109 (92–129)
Nonzero coefficients · n = 1000						
glmnet PS	Both	13 (9–19)	16 (10–25)	16 (10–26)	17 (10–28)	17 (9–29)
balnet	Treated	100 (100–100)	197 (185–209)	210 (192–230)	211 (187–242)	208 (177–246)
	Control	100 (100–100)	197 (185–209)	210 (191–230)	211 (187–242)	207 (177–246)

No MCSE, as these describe the fits. Nonzero coefficients at lambda.min: the glmnet propensity score, one fit for both arms. balnet paths ending above λ = 1.01e-4: in 10 of 10 cells, up to 10 000 (treated) and 10 000 (control) per cell.

Appendix Table 14: Nonzero coefficients of balnet and the glmnet propensity score under a misspecified propensity score, linear outcome. Median (lower–upper quartile) across replications.

		p/*n*				
		0.1	0.5	1	2	4
Nonzero coefficients · n = 500						
glmnet PS	Both	8 (5–11)	9 (5–15)	10 (6–17)	11 (6–19)	11 (6–21)
balnet	Treated	50 (50–50)	102 (95–110)	110 (99–121)	112 (98–128)	110 (93–131)
	Control	50 (50–50)	106 (99–113)	113 (103–124)	114 (101–131)	111 (95–133)
Nonzero coefficients · n = 1000						
glmnet PS	Both	9 (6–14)	11 (7–18)	11 (6–19)	12 (7–21)	12 (7–22)
balnet	Treated	100 (100–100)	201 (188–213)	213 (195–234)	216 (191–246)	212 (181–251)
	Control	100 (100–100)	207 (195–219)	219 (201–241)	220 (195–250)	216 (184–256)

No MCSE, as these describe the fits. Nonzero coefficients at lambda.min: the glmnet propensity score, one fit for both arms. balnet paths ending above λ = 1.01e-4: in 10 of 10 cells, up to 10 000 (treated) and 10 000 (control) per cell.

Appendix F: Sparsity grid

Performance

Appendix Table 15: Bias, RMSE and nonzero coefficients of the glmnet propensity score once the balnet path truncates, under correct specification.

			Bias				RMSE		Propensity score
Covariates	v	s	balnet	ABW	Hájek	AIPW	ABW	AIPW	Nonzero
independent	1	4	−0.171	−0.074	−0.329	−0.122	0.129	0.159	10–16
		16	−0.207	−0.117	−0.376	−0.186	0.158	0.212	13–21
		64	−0.223	−0.137	−0.393	−0.209	0.174	0.233	13–22
	0.5	4	−0.183	−0.075	−0.354	−0.132	0.130	0.166	12–20
		16	−0.296	−0.197	−0.490	−0.300	0.225	0.318	22–41
		64	−0.374	−0.308	−0.579	−0.447	0.328	0.460	21–52
AR(1)	1	4	−0.122	−0.039	−0.278	−0.071	0.108	0.120	7–11
		16	−0.146	−0.058	−0.327	−0.116	0.118	0.151	12–19
		64	−0.157	−0.072	−0.346	−0.140	0.125	0.171	15–22
	0.5	4	−0.126	−0.038	−0.285	−0.073	0.110	0.122	8–12
		16	−0.176	−0.076	−0.376	−0.148	0.128	0.176	17–29
		64	−0.230	−0.140	−0.466	−0.256	0.174	0.274	30–50
equicorrelated	1	4	−0.091	−0.047	−0.212	−0.059	0.111	0.119	10–22
		16	−0.062	−0.036	−0.189	−0.044	0.106	0.110	13–29
		64	−0.047	−0.028	−0.178	−0.034	0.104	0.107	20–32
	0.5	4	−0.088	−0.046	−0.214	−0.061	0.110	0.117	11–24
		16	−0.053	−0.037	−0.184	−0.046	0.107	0.109	14–33
		64	−0.029	−0.019	−0.162	−0.022	0.101	0.103	21–34

Means over the 8 cells of each row (n = 500 and 1000, p/n = 0.5 and 1, linear and quadratic outcomes), at 1000 Monte Carlo replications per cell. Largest MCSE: 0.0035 (bias), 0.0034 (RMSE). Nonzero: range of the per-cell median nonzero coefficients of the glmnet propensity score, over p/n = 0.1 to 1. ABW has the lower RMSE in all 96 cells with independent or AR(1) covariates. While exact balance holds (p/n = 0.1) the absolute bias of balnet is at most 0.006 (linear) and 0.028 (quadratic).

Imbalance bound

Appendix Table 16: Linear-outcome bias of balnet as a percentage of the imbalance bound, the sum of the two arms' mean path ends times the ℓ_1 norm of the outcome coefficient.

			n = 500, p/n		n = 1000, p/n	
s	v	Covariates	0.5	1	0.5	1
4	1.0	independent	98	97	99	100
4	1.0	AR(1)	94	99	100	98
4	0.5	independent	99	99	100	100
4	0.5	AR(1)	96	98	99	101
16	1.0	independent	78	77	85	83
16	1.0	AR(1)	79	83	86	86
16	0.5	independent	88	87	95	95
16	0.5	AR(1)	80	84	91	93
64	1.0	independent	60	60	67	66
64	1.0	AR(1)	62	63	69	70
64	0.5	independent	60	58	72	71
64	0.5	AR(1)	57	64	68	73

Largest MCSE: 2.7 points; values above 100 are within 1 MCSE of 100. The ℓ_1 norm is $\sum_{j \le s} j^{-\nu}/\sigma_g$.

Averages

Appendix Table 17: Per cent lower RMSE $\bar{R}(\Theta)$ and share of bias $\bar{s}_m(\Theta)$ of A against B at exact balance ($p/n = 0.1$).

	$\bar{R}(\Theta)$, A vs B			$\bar{s}_m(\Theta)$	
	balnet vs glmnet (Hájek)	balnet vs AIPW-glmnet	ABW-balnet vs AIPW-glmnet	balnet	glmnet (Hájek)
Overall					
all cells	55.6	6.1	6.1	0.01	0.80
Confounders s					
4	55.8	0.5	0.5	0.02	0.81
16	59.3	10.3	10.3	0.01	0.83
64	60.9	15.6	15.6	0.01	0.82
Decay ν					
0.5	57.0	7.6	7.6	0.01	0.81
1	54.1	4.6	4.6	0.02	0.79
Covariates					
independent	63.6	18.1	18.1	0.01	0.88
AR(1)	58.2	2.7	2.7	0.01	0.84
equicorrelated	42.4	-3.8	-3.8	0.02	0.69
Independent covariates, confounders s					
4	62.2	5.8	5.8	0.01	0.87
16	67.5	24.2	24.2	0.01	0.91
64	71.1	35.0	35.0	0.01	0.92
AR(1) covariates, confounders s					
4	56.5	-2.7	-2.7	0.02	0.83
16	62.6	5.4	5.4	0.01	0.88
64	65.3	12.5	12.5	0.01	0.89
Equicorrelated covariates, confounders s					
4	47.7	-1.7	-1.7	0.03	0.74
16	44.6	-0.8	-0.8	0.02	0.70
64	40.4	-5.9	-5.9	0.02	0.67

Largest MCSE: 1.2 points ($\bar{R}(\Theta)$), 0.015 ($\bar{s}_m(\Theta)$). Correct specification, linear and quadratic outcomes. The overall row is over 60 cells, 30 per outcome, at 1000 Monte Carlo replications; the rows by s use n = 1000 only (s = 64 is not run at n = 500).

Appendix Table 18: Per cent lower RMSE $\bar{R}(\Theta)$ and share of bias $\bar{s}_m(\Theta)$ of A against B once the balnet path truncates ($p/n \geq 0.5$).

	$\bar{R}(\Theta)$, A vs B			$\bar{s}_m(\Theta)$	
	balnet vs glmnet (Hájek)	balnet vs AIPW-glmnet	ABW-balnet vs AIPW-glmnet	balnet	glmnet (Hájek)
Overall					
all cells	44.2	-10.3	17.8	0.59	0.88
Confounders s					
4	44.1	-24.8	12.3	0.59	0.87
16	44.5	-8.8	19.2	0.60	0.88

	$\bar{R}(\Theta)$, A vs B			$\bar{s}_m(\Theta)$	
	balnet vs glmnet (Hájek)	balnet vs AIPW-glmnet	ABW-balnet vs AIPW-glmnet	balnet	glmnet (Hájek)
64	44.0	1.3	21.7	0.59	0.87
Decay v					
0.5	43.9	-7.0	19.3	0.61	0.88
1	44.5	-13.7	16.4	0.58	0.87
Covariates					
independent	39.5	-8.0	25.4	0.81	0.94
AR(1)	47.4	-15.9	22.9	0.69	0.92
equicorrelated	45.3	-7.2	3.5	0.28	0.76
Independent covariates, confounders s					
4	42.5	-27.8	20.1	0.73	0.92
16	39.2	-4.2	28.1	0.84	0.95
64	36.8	5.5	27.8	0.86	0.95
AR(1) covariates, confounders s					
4	46.4	-33.0	10.0	0.60	0.89
16	47.9	-17.3	24.7	0.70	0.93
64	48.0	0.2	32.5	0.76	0.94
Equicorrelated covariates, confounders s					
4	43.3	-14.3	6.1	0.45	0.81
16	46.1	-5.5	2.7	0.26	0.77
64	46.6	-2.1	1.8	0.13	0.72

Largest MCSE: 0.5 points ($\bar{R}(\Theta)$), 0.007 ($\bar{s}_m(\Theta)$). Correct specification, linear and quadratic outcomes. The overall row is over 144 cells, 72 per outcome, at 1000 Monte Carlo replications.

Tan's transformation

Appendix Table 19: RMSE under Tan's transformation, p/n = 0.1.

			RMSE, linear				RMSE, quadratic			
Covariates	*v*	*s*	balnet	ABW	Hájek	AIPW	balnet	ABW	Hájek	AIPW
n = 500										
independent	1	4	0.152	0.151	0.368	0.201	0.158	0.158	0.219	0.150
		16	0.153	0.153	0.399	0.229	0.162	0.163	0.273	0.155
	0.5	4	0.160	0.160	0.399	0.207	0.166	0.167	0.236	0.154
		16	0.166	0.165	0.471	0.245	0.177	0.178	0.277	0.121
AR(1)	1	4	0.138	0.138	0.314	0.158	0.137	0.137	0.230	0.152
		16	0.136	0.136	0.346	0.168	0.137	0.137	0.226	0.133
	0.5	4	0.128	0.128	0.318	0.155	0.134	0.134	0.217	0.140
		16	0.131	0.130	0.372	0.166	0.137	0.137	0.265	0.131
equicorrelated	1	4	0.125	0.125	0.251	0.132	0.129	0.129	0.197	0.138
		16	0.125	0.125	0.236	0.120	0.126	0.126	0.183	0.121
	0.5	4	0.136	0.136	0.256	0.137	0.130	0.130	0.188	0.126
		16	0.122	0.122	0.230	0.114	0.123	0.123	0.180	0.114
n = 1000										
independent	1	4	0.122	0.122	0.308	0.174	0.129	0.129	0.165	0.149
		16	0.133	0.133	0.353	0.209	0.135	0.135	0.204	0.137
		64	0.133	0.132	0.364	0.217	0.137	0.137	0.212	0.128
	0.5	4	0.137	0.136	0.346	0.181	0.137	0.138	0.158	0.124
		16	0.148	0.148	0.425	0.223	0.151	0.152	0.214	0.093

			RMSE, linear				RMSE, quadratic			
Covariates	v	s	balnet	ABW	Hájek	AIPW	balnet	ABW	Hájek	AIPW
		64	0.149	0.148	0.485	0.255	0.156	0.157	0.279	0.088
AR(1)	1	4	0.104	0.104	0.265	0.136	0.100	0.101	0.254	0.181
		16	0.108	0.108	0.294	0.148	0.104	0.104	0.180	0.118
		64	0.109	0.109	0.301	0.154	0.107	0.108	0.171	0.105
	0.5	4	0.103	0.103	0.262	0.132	0.101	0.101	0.170	0.128
		16	0.108	0.108	0.329	0.143	0.107	0.108	0.183	0.086
		64	0.110	0.109	0.369	0.162	0.113	0.113	0.216	0.082
equicorrelated	1	4	0.095	0.095	0.200	0.111	0.094	0.094	0.161	0.119
		16	0.090	0.090	0.189	0.094	0.089	0.089	0.152	0.097
		64	0.090	0.091	0.180	0.088	0.091	0.091	0.130	0.087
	0.5	4	0.096	0.096	0.207	0.110	0.096	0.096	0.153	0.109
		16	0.089	0.089	0.177	0.086	0.090	0.090	0.146	0.091
		64	0.090	0.090	0.169	0.082	0.092	0.092	0.137	0.086

ABW: ABW-balnet; Hájek: glmnet (Hájek); AIPW: AIPW-glmnet. Largest MCSE: 0.08 (RMSE). No estimate is missing. $\tau = 0$.

Common factor share

Appendix Table 20: Share of the propensity index's variance carried by the common factor Z_{i0} under equicorrelation, $(\sum_{j\le s} j^{-\nu})^2/\{(\sum_{j\le s} j^{-\nu})^2 + \sum_{j\le s} j^{-2\nu}\}$.

s	$\nu = 1$	$\nu = 0.5$
4	0.75	0.79
16	0.88	0.93
64	0.93	0.98

Appendix G: Outcome signal

Performance

Appendix Table 21: Bias and RMSE under correct specification, by outcome, $n = 1000$.

		Signal strength κ							
		0.02	0.05	0.1	0.2	0.35	0.5	0.7	1
Linear									
balnet	Bias	0.000	0.001	0.000	−0.001	0.000	−0.002	0.000	0.000
	RMSE	0.076	0.077	0.076	0.077	0.076	0.078	0.076	0.076
glmnet (Hájek)	Bias	−0.006	−0.014	−0.028	−0.058	−0.101	−0.146	−0.201	−0.286
	RMSE	0.068	0.070	0.074	0.090	0.122	0.163	0.215	0.298
Oracle (Hájek)	Bias	−0.001	0.001	0.000	−0.002	−0.001	−0.004	−0.004	−0.003
	RMSE	0.072	0.073	0.073	0.076	0.083	0.095	0.111	0.141
Quadratic									
balnet	Bias	−0.001	0.000	−0.002	−0.005	−0.007	−0.012	−0.014	−0.018
	RMSE	0.076	0.077	0.076	0.080	0.087	0.098	0.113	0.142
glmnet (Hájek)	Bias	−0.008	−0.017	−0.035	−0.071	−0.123	−0.177	−0.244	−0.346
	RMSE	0.069	0.071	0.077	0.102	0.147	0.201	0.269	0.376
Oracle (Hájek)	Bias	−0.001	0.001	0.000	−0.002	−0.002	−0.005	−0.004	0.000
	RMSE	0.072	0.074	0.074	0.082	0.100	0.124	0.155	0.214
Exponential									
balnet	Bias	0.000	0.000	0.000	−0.002	−0.001	−0.004	−0.002	−0.003
	RMSE	0.076	0.077	0.076	0.077	0.077	0.079	0.079	0.082

		Signal strength κ							
		0.02	0.05	0.1	0.2	0.35	0.5	0.7	1
glmnet (Hájek)	Bias	−0.006	−0.012	−0.025	−0.051	−0.087	−0.127	−0.174	−0.247
	RMSE	0.068	0.070	0.072	0.086	0.112	0.147	0.190	0.262
Oracle (Hájek)	Bias	−0.001	0.001	0.000	−0.002	−0.001	−0.003	−0.003	−0.001
	RMSE	0.072	0.073	0.072	0.075	0.081	0.091	0.103	0.130

Largest MCSE: 0.0021 (Bias), 0.0018 (RMSE). No estimate is missing. $\tau = 0$.

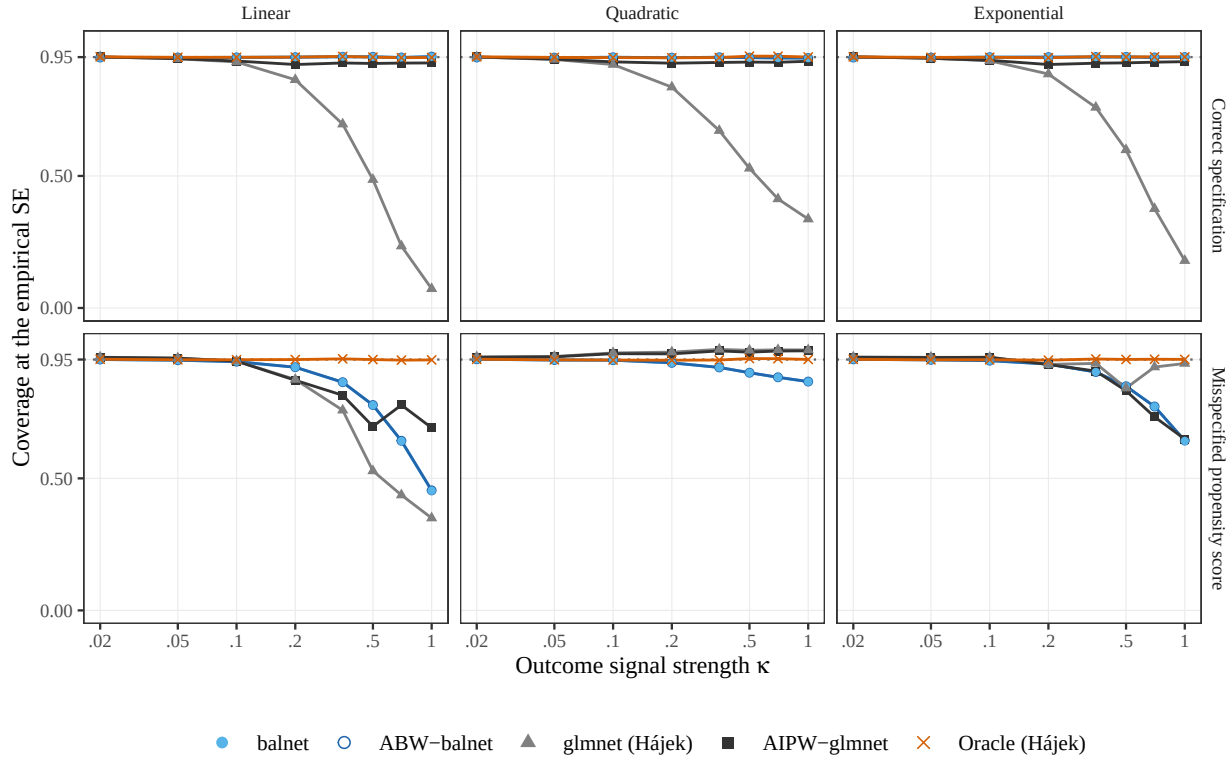
Fit diagnostics

Appendix Table 22: Effective sample size of the balnet and glmnet weights under correct specification, $n = 1000$. Median (lower–upper quartile) across replications. The weights do not depend on the outcome.

		Signal strength κ	
		0.02	0.05
ESS			
balnet	Treated	345 (326–363)	345 (326–364)
	Control	345 (326–364)	344 (325–363)
glmnet	Treated	430 (413–445)	429 (413–445)
	Control	430 (413–446)	429 (412–445)

No MCSE, as these describe the fits. ESS: Kish's, per arm. No balnet path ends above $\lambda = 1.01e\text{-}4$.

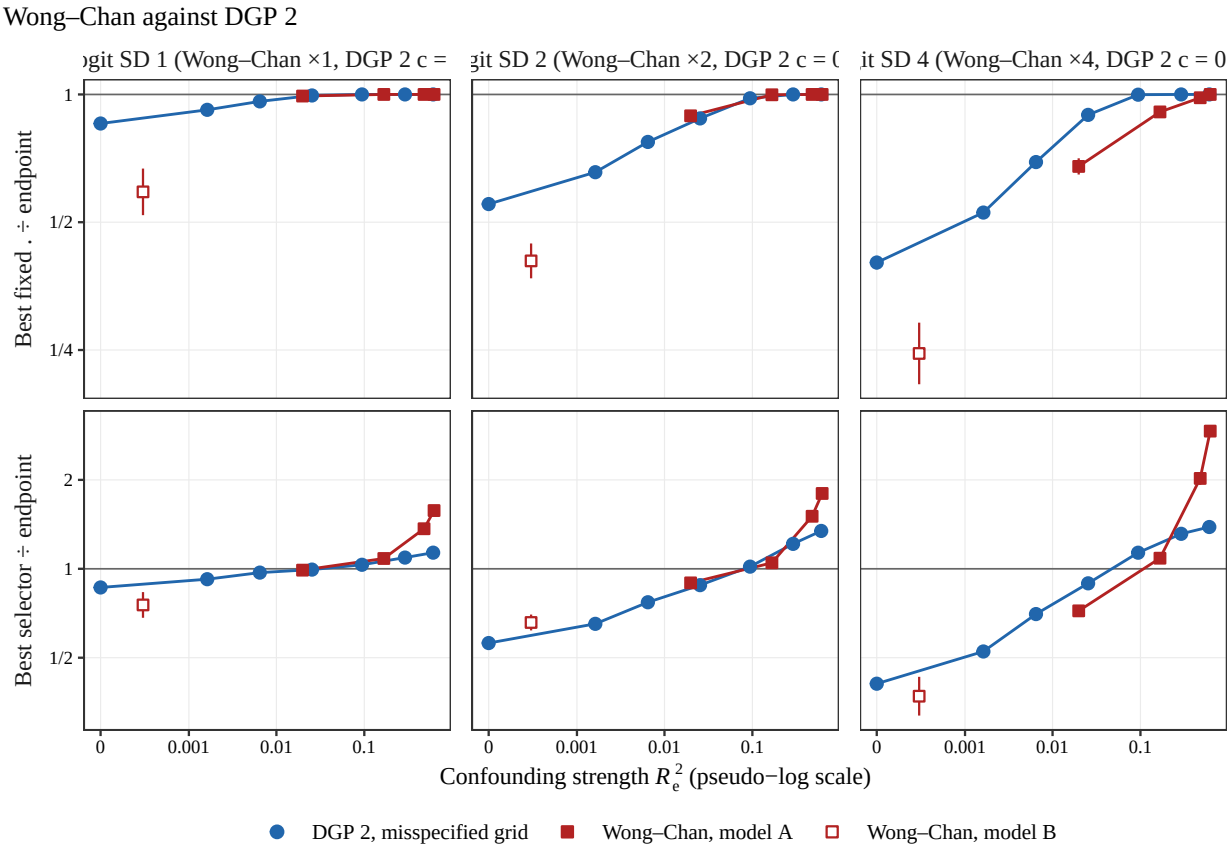
Coverage



Appendix Figure 5: Coverage at the empirical SE by outcome signal strength κ at $n = 1000$, both specifications. The dotted line is the nominal 95%.

Appendix H: Wong–Chan design

The design of Wong and Chan (2018) as run by Wang and Zubizarreta (2020) has ten standard normal covariates, a logistic treatment on two, outcome model A linear in four and model B nonlinear in them, and an analyst who sees Kang–Schafer transforms of the four. We use $n = 5000$, the ATT and 1000 replications, `balnet` balancing the covariates and their squares on a 100-point λ grid, the logit scaled by 1, 2 or 4 and the noise SD σ at 1, 10, 30 and 100. Cells match DGP 2's misspecified grid on the SD of the true logit ($c = 1$, 0.5, 0.25) and on R_e^2, the share of the control outcome's variance the true score explains, which neither overlap knob changes. The endpoint is the last λ reached by 95% of replications for Wong–Chan and each replication's own last λ for DGP 2.



Appendix Figure 6: RMSE at the best fixed λ (top) and of the best selector (bottom), relative to the endpoint's, against R_e^2 by logit SD. Wong–Chan points are model A at $\sigma = 100$, 30, 10 and 1 (left to right) and model B at $\sigma = 1$ (open), DGP 2 points $\kappa = 0$ to 1 under the linear outcome.

Appendix Table 23: Interior optima and selector wins, each Wong–Chan cell against the DGP 2 cells bracketing its R_e^2 at the same logit SD.

Matched cells (R_e^2)		Best fixed λ		Best selector		Interior optimum		Selector win	
Wong–Chan	DGP 2	W–C	DGP 2	W–C	DGP 2	W–C	DGP 2	W–C	DGP 2
Propensity logit SD 1 (Wong–Chan ×1, DGP 2 c = 1)									
B, $\sigma = 1$ (0.0003)	$\kappa = 0$ (0)	0.59	0.85	0.75	0.86	yes	yes	yes	yes
A, $\sigma = 100$ (0.020)	$\kappa = 1/16$–$1/8$ (0.0065–0.025)	0.99	0.96–0.99	0.99	0.97–0.99	no	no	no	no
A, $\sigma = 30$ (0.17)	$\kappa = 1/4$–$1/2$ (0.094–0.29)	1.00	1.00	1.08	1.03–1.09	no	no	no	no
A, $\sigma = 10$ (0.48)	$\kappa = 1/2$–1 (0.29–0.61)	1.00	1.00	1.37	1.09–1.13	no	no	no	no
A, $\sigma = 1$ (0.62)	$\kappa = 1$ (0.61)	1.00	1.00	1.57	1.13	no	no	no	no

Matched cells (R^2_e)		Best fixed λ		Best selector		Interior optimum		Selector win	
Wong–Chan	DGP 2	W–C	DGP 2	W–C	DGP 2	W–C	DGP 2	W–C	DGP 2
Propensity logit SD 2 (Wong–Chan ×2, DGP 2 c = 0.5)									
B, σ = 1 (0.0003)	κ = 0 (0)	0.41	0.55	0.66	0.56	yes	yes	yes	yes
A, σ = 100 (0.020)	κ = 1/16–1/8 (0.0065–0.025)	0.89	0.77–0.88	0.90	0.77–0.88	yes	yes	yes	yes
A, σ = 30 (0.17)	κ = 1/4–1/2 (0.094–0.29)	1.00	0.98–1.00	1.05	1.02–1.21	no	no	no	no
A, σ = 10 (0.48)	κ = 1/2–1 (0.29–0.61)	1.00	1.00	1.51	1.21–1.34	no	no	no	no
A, σ = 1 (0.62)	κ = 1 (0.61)	1.00	1.00	1.80	1.34	no	no	no	no
Propensity logit SD 4 (Wong–Chan ×4, DGP 2 c = 0.25)									
B, σ = 1 (0.0003)	κ = 0 (0)	0.25	0.40	0.37	0.41	yes	yes	yes	yes
A, σ = 100 (0.020)	κ = 1/16–1/8 (0.0065–0.025)	0.68	0.69–0.90	0.72	0.70–0.89	yes	yes	yes	yes
A, σ = 30 (0.17)	κ = 1/4–1/2 (0.094–0.29)	0.91	1.00	1.09	1.13–1.31	yes	no	no	no
A, σ = 10 (0.48)	κ = 1/2–1 (0.29–0.61)	0.98	1.00	2.02	1.31–1.39	no	no	no	no
A, σ = 1 (0.62)	κ = 1 (0.61)	1.00	1.00	2.92	1.39	no	no	no	no

Ratios: RMSE at the best fixed λ on the path, and of the best of the five shared selectors, over RMSE at the path endpoint; below 1 the endpoint is beaten. Each Wong–Chan cell is set against the cells of DGP 2's misspecified grid at the same logit SD whose R^2_e (in brackets) bracket its own, model B against κ = 0; a DGP 2 entry spans its two cells. Interior optimum and selector win as defined in Performance measures, applied to both designs. The designs agree on the interior optimum in 14 of 15 matched cells and on the selector win in 15 of 15; the exception: logit SD 4, σ = 30 (interior optimum yes against no). Largest MCSE of a log ratio 0.085 (Wong–Chan), 0.019 (DGP 2).

Appendix I: Study 2

Appendix Table 24: Selector wins, lowest RMSE and mean RMSE change by selector, all Study 2 grids.

			Selector win (%)		Win given an interior optimum (%)		Lowest RMSE (%)		Mean RMSE change (%)	
Selector	Criterion	Scoring	balnet	ABW	balnet	ABW	balnet	ABW	balnet	ABW
`cv.bloss`	`bloss`	Cross-validation	54	74	89	93	26	25	-1	-16
`cv.smd`	`smd`	Cross-validation	40	50	65	63	15	19	30	-7
`cv.inf`	`inf`	Cross-validation	43	55	70	70	16	20	16	-13
`boot.smd`	`smd`	Bootstrap	38	51	63	64	13	11	11	-11
`boot.inf`	`inf`	Bootstrap	52	70	85	89	30	25	-5	-15

Shares of the 1803 cells per estimator over the ten Study 2 grids. Selector win: RMSE below the endpoint's by more than three times their combined MCSE. Lowest RMSE: lowest of the five in the cell, whether or not it wins. Mean RMSE change: geometric mean of the selector's RMSE over the endpoint's, all cells.

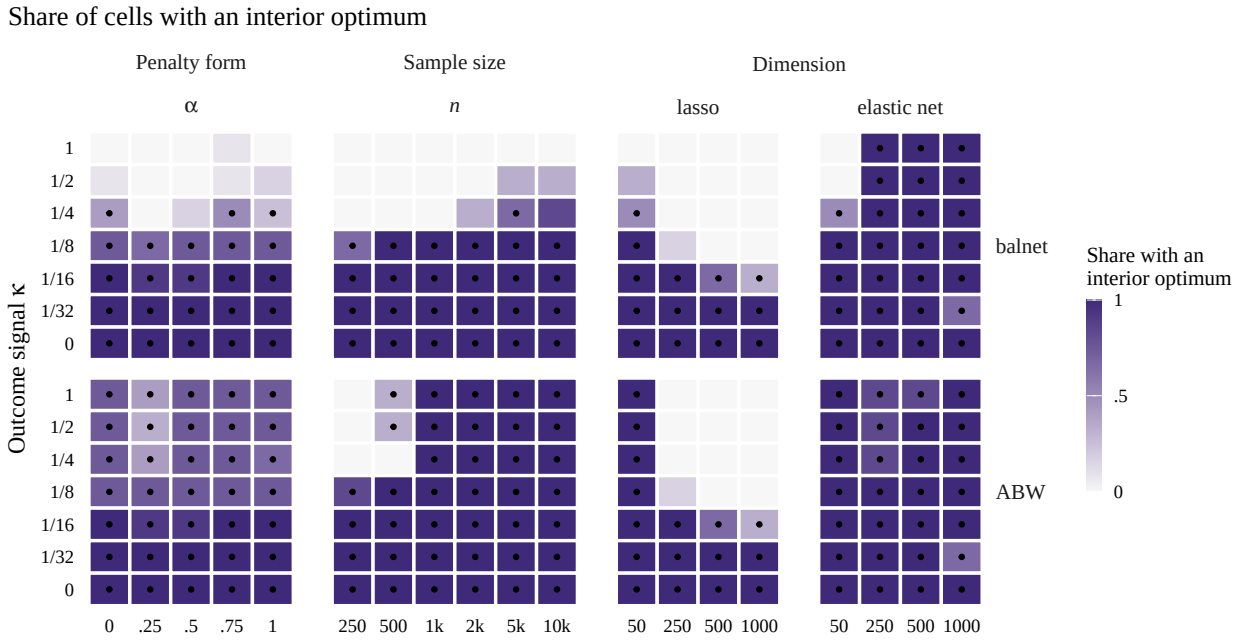
Penalty form

Appendix Table 25: Interior optima, selector wins and RMSE relative to the lasso by penalty form α.

	Interior optimum (%)		Selector win (%)		RMSE / lasso, endpoint		RMSE / lasso, optimum	
α	balnet	ABW	balnet	ABW	balnet	ABW	balnet	ABW
1 (lasso)	60	85	56	82	1.00	1.00	1.00	1.00
0.75	63	86	55	83	1.05	0.91	1.02	0.94
0.5	55	85	51	85	1.07	0.89	1.06	0.92
0.25	51	69	49	64	1.08	0.83	1.08	0.89
0 (ridge)	61	86	55	83	1.08	0.91	1.08	0.89

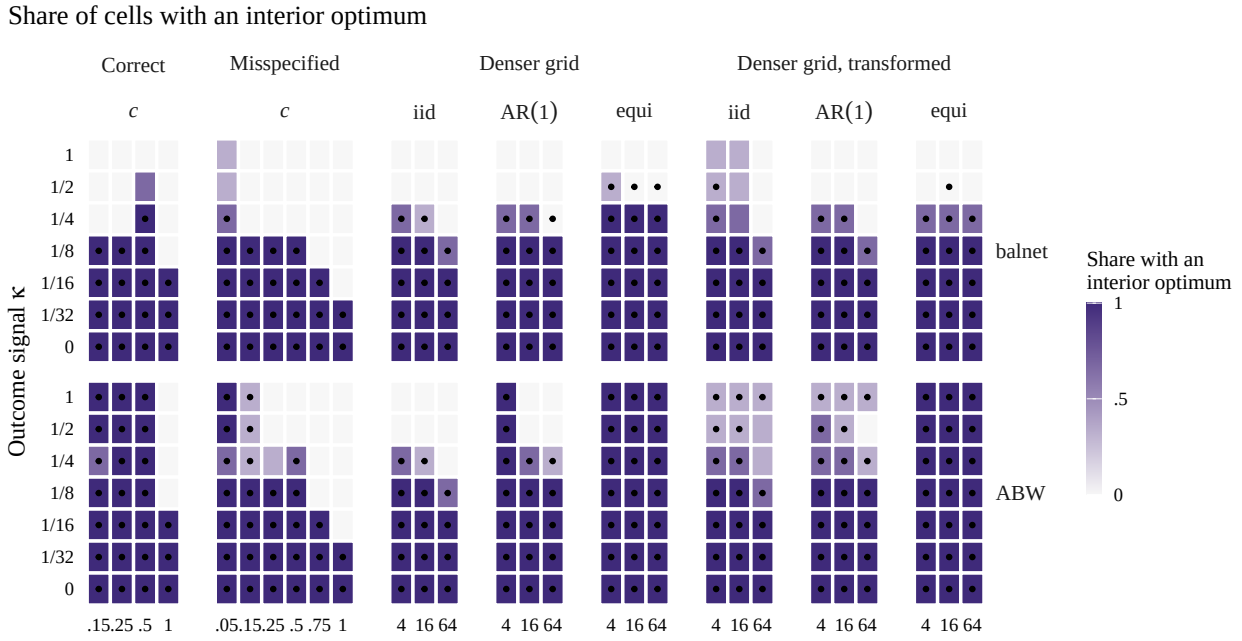
Shares over the 84 cells per α and estimator (four overlaps, seven κ, three outcomes). Ratios: geometric mean over the 18 cells at $c \leq 0.25$ and κ = 1 of the RMSE at the endpoint, or at the path minimum, to the lasso's in the same cell. The endpoints differ by construction and the forms do not share draws. With no signal every $\alpha < 1$ lowers the endpoint RMSE of both estimators by 13% to 24%, and at c = 1 every form stays within 3% of the lasso at both points.

Sample size and dimension



Appendix Figure 7: Share of cells with an interior optimum by κ and α, n or p, pooled over each grid's outcomes and overlap levels (Table 3). A dot marks a selector win in any pooled cell.

Misspecification and denser confounders



Appendix Figure 8: As Appendix Figure 7, by c under each specification and by s and covariance structure in the denser grid, with and without Tan's transformation.